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Research paper



Learning to sail without wind information: reinforcement learning-based control for unmanned sailing vessels under partial observability

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ABSTRACT

Unmanned sailing vessels commonly rely on real-time wind-field measurements as key inputs for multi-level decision-making, including path planning and control. During long-endurance offshore missions, however, wind-sensing systems may be affected by harsh environmental conditions, external mounting constraints, and maintenance difficulties. These issues motivate complementary control strategies that can support autonomous navigation while reducing dependence on real-time wind-measurement channels. In this study, we formulate sail–rudder control under partial observability, where the policy uses onboard observable vessel states and target-relative information without receiving measured wind speed or wind direction. A recurrent SAC-LSTM policy is employed as a wind-input-free control implementation, in which a Long Short-Term Memory (LSTM) module is incorporated into the Soft Actor-Critic (SAC) framework to encode recent observation–action history. This design is used to examine whether temporal context from vessel-motion evolution and previous sail–rudder actions can support continuous sail–rudder decision-making when direct wind variables are excluded from the policy input. In randomized-wind simulations evaluated over five independent training seeds, SAC-LSTM achieved a task success rate of 99.36%, compared with 39.64% for the memoryless wind-input-free SAC baseline and 95.24% for the wind-informed SAC reference. Under the same wind-input-free current-observation setting, SAC-LSTM also reduced the average successful-episode navigation time and rudder-angle variation rate relative to the memoryless SAC baseline. In the present randomized-wind benchmark, SAC-LSTM obtained a higher task success rate than the wind-informed SAC reference without using measured wind speed or wind direction, while showing similar successful-episode navigation time and trajectory length. These results indicate that recurrent observation–action history can provide useful temporal context for wind-input-free sail–rudder control under the tested 4-DOF still-water simulation conditions.

1. Introduction

Ocean observation increasingly requires persistent, large-scale, and cost-effective in situ data collection. Autonomous marine vehicles have therefore become important platforms for acquiring oceanographic data and supporting long-duration marine monitoring tasks (Chai et al., 2020; Verfuss et al., 2019). Among these platforms, unmanned sailing vessels are particularly attractive because they can exploit renewable wind energy, often in combination with solar power, to extend operational endurance and reduce dependence on conventional propulsion systems. This makes them suitable for long-range environmental monitoring and air–sea interface observation. However, the effectiveness of such platforms depends critically on reliable autonomous navigation, which directly influences mission efficiency, operational safety, and long-endurance deployment capability.

The navigation and control of unmanned sailing vessels are challenging because wind plays a dual role in their operation. On the one hand, wind is the primary propulsion source that determines the

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achievable sailing state. On the other hand, it is also a highly variable environmental factor that introduces uncertainty into vessel motion and control decision-making. Unlike waves and currents, which are commonly treated as external disturbances in many control frameworks, wind directly governs the generation of sail force and strongly affects the coupling among vessel velocity, heading, sail angle, and rudder action. Different wind conditions may require fundamentally different strategies: tailwind conditions may allow relatively direct navigation, crosswind conditions require coordinated heading and sail adjustment, whereas headwind conditions often require tacking or zigzag maneuvers. Therefore, effective point-to-point navigation requires not only feasible route generation but also real-time sail–rudder collaborative control under nonlinear dynamics and environmental uncertainty.

Most existing autonomous sailing approaches can be broadly understood within a wind-dependent guidance–navigation–control (GNC) paradigm. In such frameworks, wind measurements or wind-aware models are commonly used to support performance prediction, route planning, sail angle selection, and trajectory tracking. These methods have demonstrated strong effectiveness when environmental information is reliable, and they remain important references for autonomous sailing system design. Nevertheless, their performance may be affected by the quality and robustness of wind sensing. In practical long-endurance missions, wind sensors (e.g., Airmar) are often externally mounted and may be exposed to motion-induced measurement errors, mechanical degradation, harsh marine conditions, and maintenance difficulties. For hybrid or compact marine platforms (e.g., Ocean Aero), the installation and long-term reliability of external wind sensing devices may also become a practical constraint. These sensing constraints motivate complementary control strategies that can reduce dependence on direct wind input while preserving effective navigation performance.

This study therefore addresses the following research question: Can an unmanned sailing vessel achieve reliable point-to-point navigation while reducing reliance on direct wind measurements through the temporal evolution of onboard observable states and previous control actions? Without direct observations of wind direction and wind speed, identical or similar observations of vessel motion may correspond to different hidden wind conditions and therefore require different sail–rudder actions. The navigation problem considered in this work is therefore naturally formulated as a partially observable control problem, in which the policy must infer control-relevant latent information from historical state–action dependencies.

To address this problem, this study proposes an end-to-end reinforcement learning-based sail–rudder control framework that uses onboard observable vessel states and target-relative information as policy inputs, while treating wind as hidden environmental information. SAC is combined with an LSTM-based recurrent memory module so that recent observation–action history can be encoded together with the current observation for continuous sail–rudder decision-making. This design is used to examine whether temporal information in vessel-motion evolution and previous sail–rudder actions can support point-to-point navigation under a wind-input-free online control setting.

The main contributions of this work are summarized as follows:

(1) This study investigates wind-input-free sail–rudder control for unmanned sailing vessels under partial observability. In the proposed setting, measured wind speed and wind direction are not provided to the controller, while the policy relies on onboard vessel states, target-relative information, and recent state–action history. By incorporating recurrent sequence processing, the controller can exploit temporal dependencies in vessel motion and previous control actions, which provides useful context for decision-making under the wind-input-free observation setting.
(2) An end-to-end sail–rudder control framework is developed to generate continuous rudder and sail commands directly from observed vessel behavior. Unlike conventional wind-dependent multi-level schemes that first perform path planning or guidance and then track the planned trajectory, the proposed framework directly optimizes control behavior through interaction with the sailing dynamics. This design reduces reliance on explicit wind-based planning and tracking modules and provides a unified policy for coordinated propulsion and steering across the tested wind-direction settings.

The remainder of this paper is organized as follows. Section 2 reviews related studies on autonomous sailing vessel modeling, path planning, control design, and learning-based sail–rudder coordination, and clarifies the position of the proposed wind-input-free formulation relative to existing wind-aware approaches. Section 3 describes the sailing vessel dynamics, the time-correlated wind-field model, the observation-noise setting, and the formulation of the point-to-point navigation task under partial observability. Section 4 presents the reinforcement learning control method, including the SAC-LSTM architecture, recurrent sequence training procedure, and reward function. Section 5 reports the experimental setup, training diagnostics, comparative performance under randomized and representative wind conditions, and reward-ablation results. Section 6 discusses the main findings and limitations within the tested simulation scope. Section 7 concludes the paper.

1.1. Related work

Autonomous sailing vessel research has developed along several related directions, including vessel modeling and performance prediction, path planning, control design, and learning-based sail–rudder coordination. In most of these studies, wind-related information is used explicitly for performance prediction, route generation, sail trimming, or feedback control.

A fundamental line of research underlying autonomous sailing navigation concerns vessel modeling and numerical simulation. Dynamic models of sailing vessels, such as Jaulin's 3-DOF model (Jaulin, 2004) and Xiao's 4-DOF model (Xiao et al., 2014), provide a theoretical basis for characterizing the nonlinear and coupled dynamics of sail-equipped vessels. These models support control-oriented analysis and also serve as the foundation for simulation-based evaluation. Representative simulation platforms, including RobotX (Bingham et al., 2019) and USVSim (Paravisi et al., 2019), further enable the modeling of complex marine environments, with USVSim incorporating a comprehensive set of environmental factors encountered during real-world navigation.

In parallel, velocity prediction programs (VPPs) estimate achievable sailing performance under different wind conditions by combining environmental inputs with vessel-specific characteristics. They therefore provide quantitative references for route optimization and control design. With advances in computational techniques and hull design, velocity prediction has become increasingly accurate and adaptable (Roux et al., 2008; An et al., 2022). Together, these modeling, simulation, and velocity-prediction studies provide the foundation for evaluating vessel responses and designing navigation strategies under wind-dependent operating conditions.

Based on these modeling and simulation tools, path planning research aims to generate feasible and optimized trajectories that account for environmental constraints and vessel performance. Existing path planning studies can be broadly divided into global and local strategies. Global path planning methods operate at a larger spatial scale and are often used to determine feasible routes before local guidance and control are executed. Early studies primarily relied on classical graph-search algorithms, such as A* and Dijkstra, to generate feasible routes in discretized environments. For example, the ASV Roboat employs an A*-based planner adapted to weather forecasts (Langbein et al., 2011), while Yuan et al. (2024) improved A* using a 24-neighborhood search combined with velocity polar diagrams to enhance robustness under varying wind conditions. To better account for sailing performance, Saoud et al. integrated Dijkstra's algorithm with probabilistic

roadmaps (PRM) and velocity prediction programs (VPP), enabling optimization with respect to travel time rather than distance (Saoud et al., 2015).

In addition to deterministic search methods, bio-inspired and stochastic optimization approaches, such as ant colony optimization (ACO) and its variants, including IACO (Nian et al., 2021) and RIACO (Shen et al., 2022), have been introduced to incorporate wind-field information and environmental constraints. More recently, hybrid and learning-based strategies, including particle swarm optimization with genetic operators (Gong et al., 2020), Q-learning and Gaussian-process Q-learning (da Silva Junior et al., 2020; Liu L et al., 2022), and reinforcement learning–based planners (Suda et al., 2022), have further improved adaptability in complex and uncertain environments.

Local path planning methods, such as line-of-sight (LOS) and potential field (PF), are widely adopted due to their computational efficiency and real-time applicability. Plumet et al. (2013) proposed an LOS-based attractive potential function combined with a velocity-dependent repulsive term, while Saoud et al. (2015) and Yuan et al. (2024) enhanced PF methods by incorporating real-time VPP optimization, thereby improving adaptability in dynamic marine environments. These local strategies typically operate in conjunction with global planners to ensure both route feasibility and short-term responsiveness. Once a feasible route or local reference is generated, however, the vessel still requires a control subsystem to convert navigation objectives into coordinated sail and rudder actions.

The control subsystem is therefore responsible for executing planned trajectories through coordinated rudder and sail adjustments, corresponding to heading and speed regulation, respectively. Following the classification summarized by Tipsuwan et al. (2023), heading control methods can be broadly grouped into error-based, optimization-based, and learning-based strategies. Error-based methods, including proportional/PID controllers and line-following controllers, remain widely used because of their simplicity and ease of implementation (Jaulin et al., 2013). PID-based controllers may also be combined with gain-scheduling or fuzzy-logic mechanisms to adjust controller gains under different sailing conditions (Liu K et al., 2022). Self-steering mechanisms provide energy-saving mechanical feedback based on wind-vane actuation, but their adaptability is limited because the heading is mainly regulated with respect to an apparent-wind condition (An et al., 2021).

Optimization-based and nonlinear controllers provide stability-oriented designs for model-based sailboat control. Representative approaches include feedback linearization (Herrero et al., 2010; Wille et al., 2016), nonlinear heading/course control based on sailboat dynamics (Xiao et al., 2014; Wille et al., 2016), and robust neural/adaptive path-following extensions for unmanned robot sailboats (Zhang G. et al., 2019).

For speed regulation, sail trimming has often relied on polar-diagram or look-up-table-based strategies (Petres et al., 2011; Plumet et al., 2015), while more recent studies have introduced closed-loop or uncertainty-aware speed/path-following controllers, including adaptive nonlinear and fuzzy-logic-assisted designs (Abrougui et al., 2019; Ying-jie et al., 2019). Nevertheless, in many studies, sail control remains secondary to rudder control, which limits overall adaptability under highly dynamic wind conditions. This has motivated increasing attention to coordinated sail–rudder control, where propulsion and maneuvering are considered jointly rather than as separate subproblems.

In recent years, learning-based and optimization-based methods have received increasing attention for integrated sail–rudder decision-making. Coordinated rudder–sail control strategies based on model predictive control (MPC) (Liu et al., 2023) and deep reinforcement learning (Chen et al., 2024) have shown potential for balancing propulsion, maneuverability, and stability. These studies move beyond treating sail and rudder regulation as separate subproblems and provide useful references for integrated control design. However, they still commonly rely on explicit wind information, wind-aware models, or environment-dependent prior knowledge, which differs from the wind-input-free setting considered in this study.

Related optimization problems have also been investigated in rotor-assisted marine vehicles, another type of wind-assisted unmanned marine platform. Li et al. (2025) studied propeller thrust-saving adaptive control through a triggering-guidance mechanism, while Zhang et al. (2026) addressed robust event-triggered path following under output constraints and stochastic marine disturbances. These studies reflect common engineering concerns in wind-assisted autonomous navigation, including energy-saving operation, guidance-control coordination, and disturbance rejection. Nevertheless, rotor-assisted vehicles obtain wind-assisted propulsion through rotor-induced aerodynamic forces, whereas the present study focuses on a sail-equipped unmanned sailing vessel in which a physical sail interacts with the wind field and must be coordinated with the rudder. Therefore, these rotor-assisted studies are not direct baselines for the present sail–rudder problem, but they provide useful references for discussing energy-aware operation, guidance–control coordination, and disturbance rejection in wind-assisted unmanned marine platforms.

Overall, existing studies have established a solid foundation for autonomous sailing and wind-assisted marine vehicle control, including wind-aware modeling, VPP-based performance prediction, global and local path planning, nonlinear control, and coordinated propulsion/control regulation. These methods improve navigation performance by using environmental information, vessel models, or explicit guidance-control structures. Accordingly, the present study examines a complementary control setting in which the online policy relies on onboard observable vessel states, target-relative information, and recent observation–action history without direct wind input.

2. Modeling and problem formulation

2.1. Sailing vessel dynamics

In this study, a 4-DOF nonlinear sailing vessel model is adopted to construct a control-oriented simulation environment for point-to-point navigation. Following Xiao et al. (2014), the model describes the coupled surge, sway, roll, and yaw dynamics of the vessel under a still-water simulation setting. This formulation retains the main dynamic interactions among the sail, rudder, keel, and hull that are required for control-oriented evaluation. The calculation procedure can be organized into three steps: transforming the environmental wind into the apparent wind acting on the sail, computing the sail- and rudder-induced forces and moments, and integrating the resulting vessel accelerations to update the vessel state. The main equations used in the simulation are summarized below.

The dynamic model is formulated using an inertial reference frame

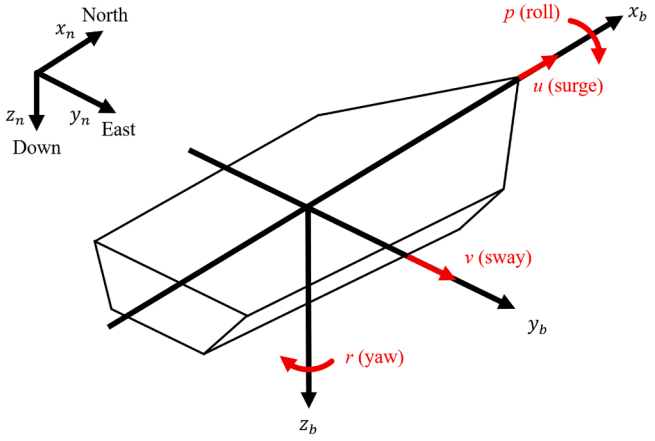


Fig. 1. 4-DOF sailing vessel model.

(n-frame) and a body-fixed reference frame (b-frame), as illustrated in Fig. 1. The generalized velocity vector in the body-fixed frame is defined as $\boldsymbol{\nu} = [u, v, p, r]^T$, where u and v are the surge and sway velocities, and p and r are the roll and yaw rates. The generalized position vector is defined as $\boldsymbol{\eta} = [x, y, \phi, \psi]^T$, where x and y denote the vessel position in the inertial frame, and ϕ and ψ denote the roll and yaw angles, respectively. The sail δ_s and rudder angles δ_r, are used as control-surface inputs for computing the corresponding aerodynamic and hydrodynamic forces and moments.

The scalar true-wind speed v_w, specified at the reference height h_1, is converted into the wind-velocity vector v_{tw} at the sail's center of effort using the absolute wind direction α_w and a logarithmic wind-profile correction determined by the effective height $|z_s|\cos\phi$ and the sea-surface aerodynamic roughness length h_0. The calculation process is as follows:

$$v_{tw} = \frac{\ln\left(\frac{|z_s|\cos\phi}{h_0}\right)}{\ln\left(\frac{h_1}{h_0}\right)}[v_w\cos\alpha_w, v_w\sin\alpha_w, 0]^T. \tag{1}$$

The wind velocity vector is then transformed into the body-fixed frame using rotation matrices associated with the vessel yaw and roll angles. The rotation matrices are defined as:

$$R_1 = \begin{bmatrix} \cos(-\psi) & -\sin(-\psi) & 0 \\ \sin(-\psi) & \cos(-\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{2}$$

$$R_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(-\phi) & -\sin(-\phi) \\ 0 & \sin(-\phi) & \cos(-\phi) \end{bmatrix} \tag{3}$$

The wind velocity in the b-frame v_{twb} is obtained as:

$$v_{twb} = R_2 R_1 v_{tw}. \tag{4}$$

The velocity of the sailboat u and v are subtracted from the wind velocity v_{twb} to calculate the apparent wind velocity v_{awb}:

$$v_{awb} = \begin{bmatrix} v_{awu} \\ v_{awv} \\ 0 \end{bmatrix} = v_{twb} - \begin{bmatrix} u \\ v \\ 0 \end{bmatrix} - \begin{bmatrix} p \\ 0 \\ r \end{bmatrix} \times \begin{bmatrix} x_s \\ y_s \\ z_s \end{bmatrix} \tag{5}$$

where p and r are the roll and yaw rates of the vessel, and x_s, y_s, z_s denote the position of the center of effort. From this, the longitudinal and lateral components of the apparent wind velocity are extracted as v_{awu} and v_{awv}. The angle of attack α_{aw} of the apparent wind is calculated as:

$$\alpha_{aw} = \arctan\left(\frac{v_{awv}}{-v_{awu}}\right). \tag{6}$$

The lift coefficient C_{ls} and drag coefficient C_{ds} can be obtained by looking up the table with the sail coefficient function based on $\alpha_{aw} - \delta_s$. The lift L_s and drag D_s forces on the sail are calculated as:

$$L_s = \frac{1}{2}\rho_a A_s\left(v_{awu}^2 + v_{awv}^2\right)C_{ls} \tag{7}$$

$$D_s = \frac{1}{2}\rho_a A_s\left(v_{awu}^2 + v_{awv}^2\right)C_{ds} \tag{8}$$

where ρ_a is the air density, A_s is the sail area.

The torque τ_{sail}, which represents the moments generated by the lift and drag forces on the sail, is computed. The components of the torque are given by:

$$\tau_{sail} = \begin{bmatrix} L_s\sin\alpha_{aw} - D_s\cos\alpha_{aw} \\ L_s\cos\alpha_{aw} + D_s\sin\alpha_{aw} \\ (L_s\cos\alpha_{aw} + D_s\sin\alpha_{aw})z_s \\ -(L_s\sin\alpha_{aw} - D_s\cos\alpha_{aw})x_{ce}\sin\delta_s \\ +(L_s\cos\alpha_{aw} + D_s\sin\alpha_{aw})(x_m - x_{ce}\cos\delta_s) \end{bmatrix} \tag{9}$$

where x_{ce} is the center of effort's x-coordinate, x_m is the position of the center of mass.

After the sail-generated aerodynamic forces and moments are obtained, the rudder-induced hydrodynamic forces are computed to describe the steering response of the vessel. The forces and moments exerted on the rudder by the relative water flow are computed in a manner analogous to those acting on the sail. The relative velocity components in the direction of the rudder motion are first computed as follows:

$$v_{aru} = -u + r{\cdot}y_r \tag{10}$$

$$v_{arv} = -v - r{\cdot}x_r + p{\cdot}z_r \tag{11}$$

where x_r, y_r and z_r represent the horizontal, lateral, and vertical positions of the rudder's center of effort relative to the center of mass of the boat. The angle of attack α_{ar} of the rudder is calculated using the relative velocity components:

$$\alpha_{ar} = \arctan\left(\frac{v_{arv}}{-v_{aru}}\right). \tag{12}$$

The rudder lift and drag coefficients C_{lr}, C_{dr} are obtained from the corresponding coefficient table as functions of the rudder angle of attack, following the same procedure used for the sail. The drag coefficient is then corrected to account for lift-induced effects, and the resulting lift and drag forces are calculated as follows:

$$C_{dr} = C_{dr} + \frac{C_{lr}^2 A_r}{2\pi\zeta_r d_r^2} \tag{13}$$

$$L_r = \frac{1}{2}\rho_w A_r\left(v_{aru}^2 + v_{arv}^2\right)C_{lr} \tag{14}$$

$$D_r = \frac{1}{2}\rho_w A_r\left(v_{aru}^2 + v_{arv}^2\right)C_{dr} \tag{15}$$

where A_r is the rudder area, ζ_r is the rudder efficiency factor, d_r is the rudder draft, and ρ_w is the water density. The torque τ_{rudder}, which consists of the moments generated by the lift and drag forces on the rudder, is calculated. The components of the torque are expressed as:

$$\tau_{rudder} = \begin{bmatrix} L_r\sin\alpha_{ar} - D_r\cos\alpha_{ar} \\ L_r\cos\alpha_{ar} + D_r\sin\alpha_{ar} \\ -(L_r\cos\alpha_{ar} + D_r\sin\alpha_{ar})\cdot z_r \\ (L_r\cos\alpha_{ar} + D_r\sin\alpha_{ar})\cdot x_r \end{bmatrix}. \tag{16}$$

The computed acceleration of the vessel in the body-fixed frame is:

$$\dot{\boldsymbol{\nu}} = -M^{-1}[(C_{RB} + C_A)\boldsymbol{\nu} + D + G] + M^{-1}(\tau_{sail} + \tau_{rudder}) \tag{17}$$

where M is the system inertia matrix, C_{RB} and C_A represent the rigid-body and added-mass Coriolis–centripetal matrices, respectively, D is the hydrodynamic damping vector, and G corresponds to the restoring force vector due to gravity and buoyancy. The time derivative of the generalized position vector in the inertial frame is then given by:

$$\dot{\boldsymbol{\eta}} = J\nu, J = \begin{bmatrix} \cos\psi & -\sin\psi\cos\phi & 0 & 0 \\ \sin\psi & \cos\psi\cos\phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \cos\phi \end{bmatrix}. \tag{18}$$

Since each output of the calculation represents the derivative of the corresponding state variable, a high-order Runge–Kutta method is used for numerical integration. The resulting dynamic equations show that the wind field affects vessel motion through the apparent wind and the resulting sail-generated forces, while the rudder and hull hydrodynamics jointly determine the maneuvering response. Therefore, wind remains an essential environmental input to the simulator, even though direct wind variables are excluded from the policy observation in the wind-input-free control setting introduced in the following subsection.

2.2. Wind field and observation setting

Building on the 4-DOF sailing vessel model, the simulation environment is constructed under still-water conditions, with wind treated as the primary environmental factor affecting sail-generated propulsion. This setting is used to examine point-to-point navigation under wind-driven partial observability, while wave-induced loads and ocean currents are not explicitly modeled.

In this study, the wind field is modeled as part of the simulation environment rather than as a policy input. The vessel therefore still operates in a wind-driven environment: the wind determines the apparent wind and the resulting sail-generated forces, but the controller is not provided with measured wind speed or wind direction. To generate time-correlated hidden wind disturbances during point-to-point navigation, a two-dimensional Dryden turbulence model is adopted.

The Dryden model is characterized by rational power spectral density functions and can capture the frequency-domain characteristics of atmospheric turbulence. Compared with simplified stochastic processes such as the Ornstein–Uhlenbeck process, the Dryden model provides a physically motivated way to generate temporally correlated wind fluctuations.

The wind field is modeled as a two-dimensional horizontal vector composed of a steady mean component and longitudinal/lateral turbulent fluctuations. Let $\bar{V}_w$ and $\bar{\psi}_w$ denote the mean wind speed and the initial mean wind direction, respectively. The turbulent components η_u and η_v, defined along and perpendicular to the mean wind direction, are generated using first-order Dryden filters:

$$\dot{\eta}_i(t) = -\frac{\bar{V}_w}{L_i}\eta_i(t) + \sqrt{\frac{2\bar{V}_w}{L_i}}\sigma_i\xi_i(t), i \in \{u, v\}, \tag{19}$$

where L_i is the turbulence length scale, σ_i is the turbulence intensity, and $\xi_i(t)$ is zero-mean unit white noise. In the discrete-time simulation, the exact first-order update is used:

$$\eta_i(k+1) = \rho_i\eta_i(k) + \sigma_i\sqrt{1-\rho_i^2}\epsilon_i(k), \tag{20}$$

$$\rho_i = \exp\left(-\frac{\Delta t}{\tau_i}\right), \tau_i = \frac{L_i}{\bar{V}_w}, \epsilon_i(k) \sim N(0,1). \tag{21}$$

The resulting global wind vector is obtained as,

$$\mathbf{V}_w(k) = \bar{V}_w\begin{bmatrix}\cos\bar{\psi}_w \\ \sin\bar{\psi}_w\end{bmatrix} + \begin{bmatrix}\cos\bar{\psi}_w & -\sin\bar{\psi}_w \\ \sin\bar{\psi}_w & \cos\bar{\psi}_w\end{bmatrix}\begin{bmatrix}\eta_u(k) \\ \eta_v(k)\end{bmatrix}. \tag{22}$$

The instantaneous wind speed and wind direction are then computed by,

$$V_w(k) = \| \mathbf{V}_w(k) \|, \psi_w(k) = atan2\left(V_{w,y}(k), V_{w,x}(k)\right). \tag{23}$$

At control step t, the corresponding wind speed and wind direction are denoted by $V_{w,t}$ and $\psi_{w,t}$, respectively. For consistency with the aerodynamic formulation in Section 3.1, $V_{w,t}$ is used as the scalar wind speed $v_{w,t}$, and $\psi_{w,t}$ corresponds to the absolute wind direction angle $\alpha_{w,t}$ in Eq. (1), i.e., $\alpha_{w,t} \equiv \psi_{w,t}$. A lower bound $V_{w,t} \geq 0.1$ m/s is imposed to avoid numerical instability in the aerodynamic calculation under extremely low wind speed. In the simulations, $\bar{V}_w = 8.0$ m/s, $L_u = 300$ m, $L_v = 150$ m, $\sigma_u = 1.2$ m/s, $\sigma_v = 1.0$ m/s, and $\Delta t = 1$ s. The initial mean wind direction $\bar{\psi}_w$ is randomly sampled from $[-\pi, \pi]$, unless a representative wind direction is specified for controlled evaluation.

To check the temporal characteristics of the selected wind-field setting, the simulated wind-direction sequence is compared with a measured wind-direction record, as shown in Figs. 2 and 3. This comparison is not intended as a site-specific wind-field validation; rather, it is used to confirm that the Dryden-generated sequence exhibits a plausible fluctuation pattern and a directional variation range comparable to the measured record. The resulting Dryden sequence is therefore adopted as a physically motivated wind disturbance input for the control-oriented simulations.

In addition to the time-varying wind field, observation noise is introduced to emulate imperfect onboard sensing within the simulation environment. These perturbations are applied only to the policy observations and not to the internal dynamic state used by the numerical simulator. For the wind-input-free policies, zero-mean Gaussian noise is added to onboard observable vessel states, including position, attitude/heading angles, linear velocities, and angular velocities. The nominal noise magnitudes are set to 1 m for position, 1° for attitude and heading angles, 0.1 m/s for linear velocities, and 0.1°/s for angular velocities.

For the wind-informed learning reference, noisy wind measurements are additionally constructed as

$$\hat{V}_{w,t} = V_{w,t} + \epsilon_{V,t}, \hat{\psi}_{w,t} = \mathrm{wrap}_{[-\pi,\pi]}\left(\psi_{w,t} + \epsilon_{\psi,t}\right), \tag{24}$$

where $\epsilon_{V,t}$ and $\epsilon_{\psi,t}$ denote zero-mean Gaussian perturbations with nominal magnitudes of 0.1 m/s and 1°, respectively. These noisy wind measurements are used only by the SAC-wind baseline that explicitly receives wind input. They are not included in the proposed SAC-LSTM observation. The proposed policy receives neither the true wind variables $(\psi_{w,t}, V_{w,t})$ nor their noisy measurements $(\hat{\psi}_{w,t}, \hat{V}_{w,t})$. This setting allows the learned policies to be evaluated under moderate sensing

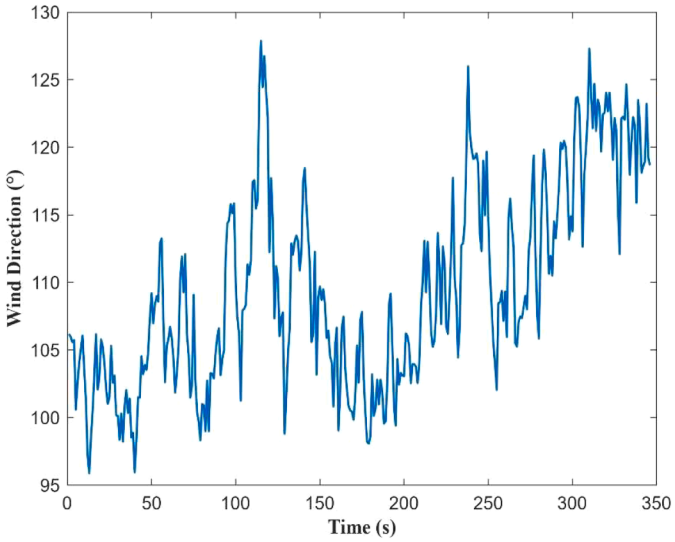


Fig. 2. Measured wind-direction record.

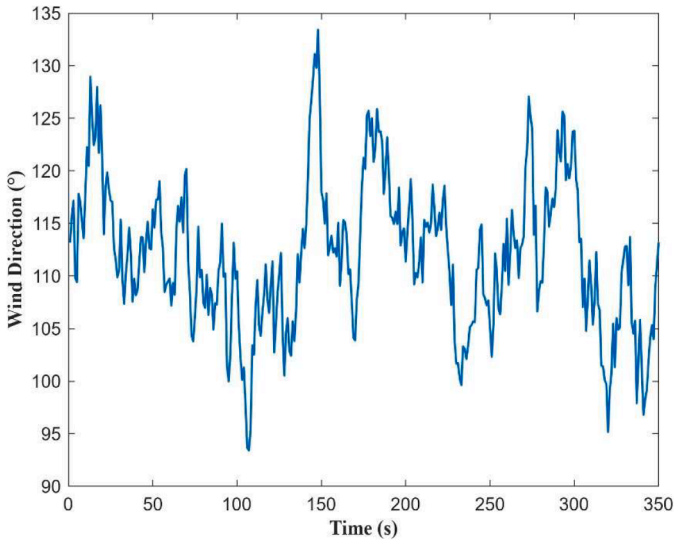


Fig. 3. Dryden-simulated wind direction.

imperfections while keeping the proposed SAC-LSTM policy free from direct wind-speed and wind-direction input.

2.3. Problem formulation under partial observability

Based on the sailing vessel dynamics and wind-field model described above, the point-to-point navigation task is formulated as a wind-input-free partially observable control problem. Two levels of state information are distinguished in this formulation. The first is the internal simulator state, which is used for numerical integration and force calculation. The second is the policy observation, which contains only the information available to the controller. At each control step t, the simulator maintains an internal state χ_t, which contains the vessel state, the current sail and rudder angles, the target position, and the environmental wind variables:

$$\chi_t = \left[x_t, y_t, \phi_t, \psi_t, u_t, v_t, p_t, r_t, \delta_{s,t}, \delta_{r,t}, x_f, y_f, \psi_{w,t}, V_{w,t}\right]. \tag{25}$$

Here, x_t and y_t denote the vessel position in the inertial frame; ϕ_t and ψ_t denote the roll angle and heading angle, respectively; u_t and v_t are the body-fixed surge and sway velocities; p_t and r_t are the roll and yaw angular velocities; $\delta_{s,t}$ and $\delta_{r,t}$ are the current sail and rudder angles; $\left(x_f, y_f\right)$ denotes the fixed target position; and $\psi_{w,t}$ and $V_{w,t}$ denote the wind direction and wind speed, respectively.

The wind variables are used by the simulator to compute the apparent wind and the sail-generated forces, but they are restricted from the wind-input-free policy observation defined below. The wind state is summarized as:

$$w_t = \left[\psi_{w,t}, V_{w,t}\right]. \tag{26}$$

The proposed SAC-LSTM policy receives a wind-input-free observation o_t. In the implementation used in this study, o_t is defined as a 14-dimensional vector:

$$o_t = \left[x_t, y_t, \phi_t, \psi_t, u_t, v_t, p_t, r_t, \delta_{s,t}, \delta_{r,t}, x_f, y_f, \Delta L_t, \Delta l_t\right]. \tag{27}$$

This observation includes onboard observable vessel states, the current sail and rudder angles, target-related information, and two derived progress features. The progress-related quantities ΔL_t and Δl_t are computed from the vessel position and the fixed target position. They provide local navigation-progress information consistent with the reward design, but they do not contain direct wind measurements. Their mathematical definitions are given in Section 4.3. For notational simplicity, the variables in o_t denote the values available to the policy after the observation-noise setting described in Section 3.2 is applied, whereas χ_t denotes the internal simulator state used for numerical integration.

For comparison, the SAC-wind baseline uses an augmented observation:

$$o_t^{wind} = \left[o_t, \widehat{\psi}_{w,t}, \widehat{V}_{w,t}\right], \tag{28}$$

where $\widehat{\psi}_{w,t}$ and $\widehat{V}_{w,t}$ denote the noisy wind measurements defined in Section 3.2. This baseline is used as a wind-informed learning reference because it explicitly receives wind information and, as described in Section 4.3, also uses a physically motivated wind-direction penalty during learning. In contrast, the proposed SAC-LSTM policy receives neither the true wind variables nor their noisy measurements. This distinction allows the wind-input-free recurrent policy to be evaluated against both a memoryless wind-input-free SAC baseline and a wind-informed SAC reference.

The control action is defined as the incremental sail–rudder command at each control step:

$$a_t = \left[\Delta\delta_{s,t}, \Delta\delta_{r,t}\right], \tag{29}$$

where $\Delta\delta_{s,t}$ and $\Delta\delta_{r,t}$ denote the incremental sail and rudder commands, respectively. After the action is applied, the simulator updates the vessel state according to the nonlinear sailing dynamics and the time-varying wind field described in Sections 3.1 and 3.2. Thus, the next internal state depends on both the commanded sail–rudder adjustment and the hidden wind-driven transition of the environment.

Under this setting, the control problem can be described as a partially observable Markov decision process:

$$M = (\mathscr{X}, \mathscr{O}, \mathscr{A}, \mathscr{P}, R, \gamma), \tag{30}$$

where $\mathscr{X}$ is the full simulator state space, $\mathscr{O}$ is the observation space available to the policy, $\mathscr{A}$ is the continuous action space, $\mathscr{P}(\chi_{t+1}|\chi_t, a_t)$ is the transition probability induced by the vessel dynamics and wind-field evolution, R is the reward function, and γ is the discount factor. The reward function is defined in Section 4.3 to guide the vessel toward the target while encouraging feasible and smooth sail–rudder control behavior.

The partial observability arises because the observation o_t does not uniquely determine the full simulator state χ_t. In particular, similar observations of vessel position, attitude, velocity, control-surface angles, and target-related quantities may correspond to different hidden wind conditions. For example, similar onboard observable vessel states may occur under tailwind, crosswind, or headwind conditions, while the corresponding feasible sail–rudder strategies can differ substantially. As a result, a memoryless policy based only on the current observation may receive insufficient information for consistent decision-making across varying wind conditions.

Although direct wind speed and wind direction are not included in the policy input, their effects are indirectly reflected in the short-term motion response of the vessel under recent sail and rudder commands. This indirect relationship does not allow the wind field to be explicitly reconstructed, but it can provide useful temporal context for control. Therefore, the control method introduced in Section 4 incorporates recent observation–action history, allowing the policy to condition its sail–rudder decisions on both the current onboard observation and the recent evolution of vessel behavior.

3. Reinforcement learning control method

Building on the wind-input-free POMDP formulation introduced in Section 3.3, this section describes the recurrent SAC implementation used for continuous sail–rudder control. The overall training and environment-interaction framework is shown in Fig. 4. During online

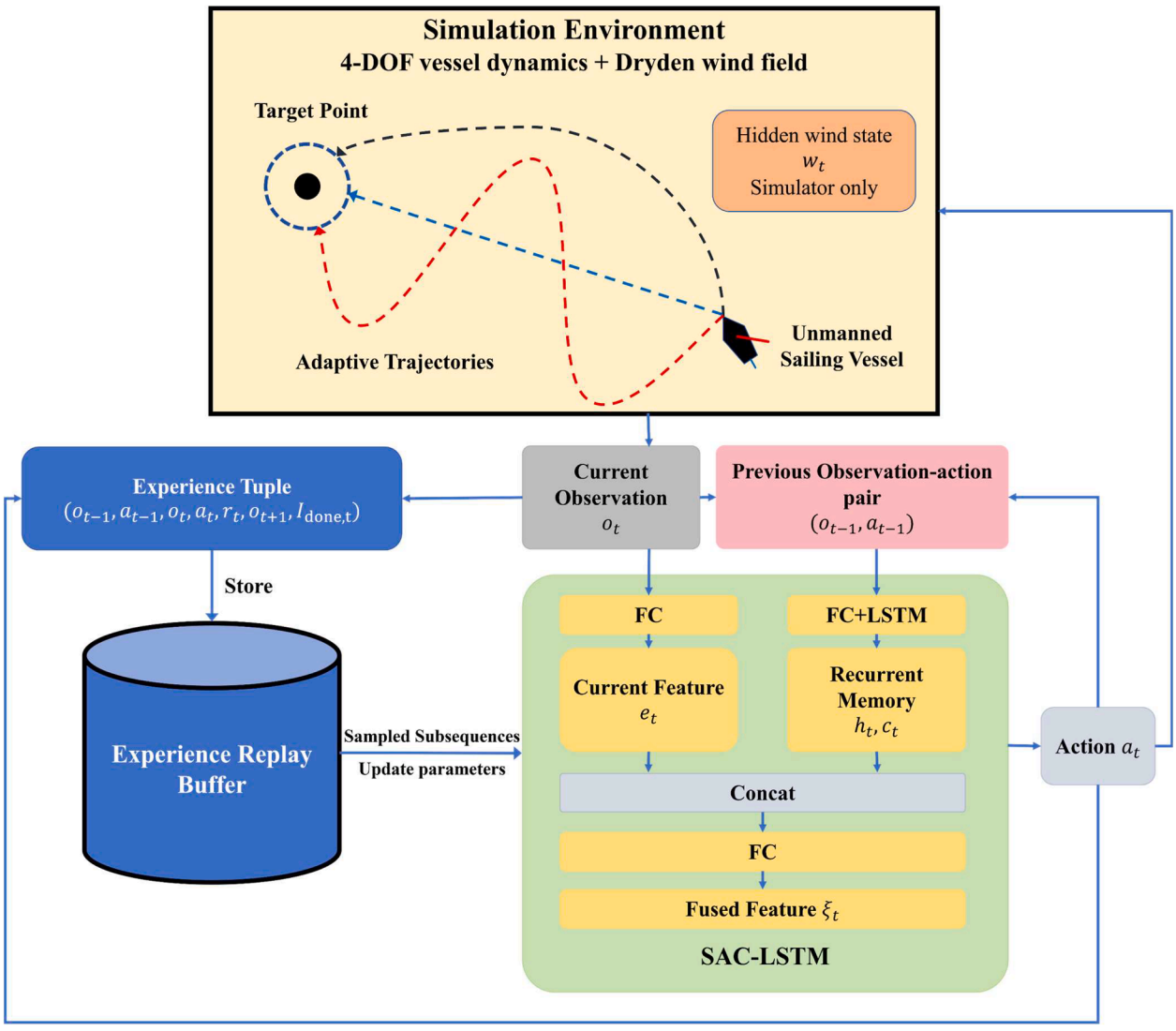


Fig. 4. Framework of model training and environment interaction under the wind-input-free partially observable setting.

interaction, the agent receives the available policy observation, propagates its recurrent memory state, and outputs incremental sail and rudder commands. The policy is then updated through off-policy SAC training using trajectory segments collected from the nonlinear sailing environment with hidden time-varying wind.

3.1. SAC-LSTM architecture

Soft Actor-Critic (SAC) is adopted as the reinforcement learning backbone for continuous sail–rudder control because it is well suited to continuous action spaces and encourages exploration through maximum-entropy policy optimization (Haarnoja et al., 2019). In the present wind-input-free setting, however, the current observation alone may not contain sufficient information for consistent control because wind variables are hidden from the policy. To provide temporal context under this partial observability, an LSTM module (Hochreiter and Schmidhuber, 1997) is incorporated into the SAC actor–critic structure as a recurrent memory component. The resulting SAC-LSTM implementation uses recent observation–action history together with the current observation to generate continuous sail and rudder commands.

To provide temporal context for decision-making, the SAC-LSTM architecture uses a recurrent branch to encode previous observation–action information. For $t \geq 1$, let o_{t-1} and a_{t-1} denote the observation and action from the previous control step, respectively. The recurrent input at step t is constructed by concatenating these two elements along the feature dimension:

$$z_t = [o_{t-1}; a_{t-1}]. \tag{31}$$

Here, z_t represents the input used to update the recurrent memory before the current action is selected. In Fig. 6, this input is visually shown as the pair (o_{t-1}, a_{t-1}) to emphasize that the recurrent branch uses the previous observation and previous action, whereas the implemented network input is their concatenated vector representation. The LSTM module updates its hidden state and cell state according to

$$(h_t, c_t) = f_{\text{LSTM}}(z_t, h_{t-1}, c_{t-1}). \tag{32}$$

In this formulation, h_t summarizes the observation–action history available before making the current decision. At the first control step of an episode, the recurrent states are initialized to zero, and the actor uses this zero-initialized memory representation for the initial action.

The current observation o_t is not directly fed into the LSTM branch at the same time step. Instead, it is processed through a separate current-observation branch for the current actor decision. For the actor network, the current observation is first processed by a feedforward feature extraction branch:

$$e_t^{\pi} = \phi_o^{\pi}(o_t), \tag{33}$$

where e_t^{π} denotes the current-observation feature used by the actor. This feature is then fused with the recurrent memory representation h_t:

$$\xi_t^{\pi} = \phi_c^{\pi}\left(\left[e_t^{\pi}; h_t\right]\right). \tag{34}$$

The actor generates a stochastic continuous control policy according to.

$$a_t \sim \pi_\theta\left(\cdot \middle| \xi_t^{\pi}\right), \tag{35}$$

where θ denotes the actor parameters. Equivalently, the actor can be understood as using the current observation and the recurrent memory representation, i.e., $\pi_\theta(\cdot|o_t, h_t)$.

For the critic networks, the same recurrent memory representation h_t is used, while the current branch evaluates a specific observation–action pair (o_t, a_t). The current observation and current action are therefore processed together through a separate feedforward branch:

$$e_t^{Q} = \phi_o^{Q}([o_t; a_t]), \tag{36}$$

where e_t^{Q} denotes the current observation–action feature. This feature is fused with the same recurrent memory representation h_t:

$$\xi_t^{Q} = \phi_c^{Q}\left(\left[e_t^{Q}; h_t\right]\right). \tag{37}$$

The twin soft Q-functions are then expressed as.

$$Q_{\omega_i}\left(\xi_t^{Q}\right), i = 1, 2, \tag{38}$$

or equivalently,

$$Q_{\omega_i}(o_t, a_t, h_t), i = 1, 2, \tag{39}$$

where ω_i denotes the parameters of the i-th critic network. This separation between actor and critic inputs follows the actor–critic structure: the actor selects an action from the current observation and memory representation, whereas the critic evaluates a specific current observation–action pair together with the same recurrent memory.

Following the standard SAC formulation, the actor is trained by minimizing the maximum-entropy policy objective:

$$J_\pi(\theta) = E\left[\alpha \log \pi_\theta\left(a_t \middle| \xi_t^{\pi}\right) - \min_{i=1,2} Q_{\omega_i}(o_t, a_t, h_t)\right], \tag{40}$$

where $a_t \sim \pi_\theta\left(\cdot \middle| \xi_t^{\pi}\right)$, and α is the temperature coefficient that balances entropy maximization and reward optimization.

The critic networks are trained by minimizing the soft Bellman residual:

$$J_Q(\omega_i) = E\left[\frac{1}{2}\left(Q_{\omega_i}(o_t, a_t, h_t) - y_t\right)^2\right], i = 1, 2. \tag{41}$$

For the target computation, the recurrent memory is advanced by one step using the current observation–action pair, so that the target critics evaluate the next observation under the corresponding next memory state. The next recurrent input is constructed as

$$z_{t+1} = [o_t; a_t], \tag{42}$$

and the next recurrent states are obtained as.

$$(h_{t+1}, c_{t+1}) = f_{\mathrm{LSTM}}(z_{t+1}, h_t, c_t). \tag{43}$$

The target value is then defined as.

$$y_t = r_t + \gamma\left(1 - I_{\mathrm{done},t}\right)\left[\min_{i=1,2} Q_{\bar{\omega}_i}(o_{t+1}, a_{t+1}, h_{t+1}) - \alpha \log \pi_\theta\left(a_{t+1} \middle| \xi_{t+1}^{\pi}\right)\right], \tag{44}$$

where $I_{\mathrm{done},t}$ is the terminal indicator, γ is the discount factor, $\bar{\omega}_i$ denotes the target critic parameters, and.

$$a_{t+1} \sim \pi_\theta\left(\cdot \middle| \xi_{t+1}^{\pi}\right). \tag{45}$$

Here, ξ_{t+1}^{π} is constructed from the next observation o_{t+1} and the next recurrent memory h_{t+1}. The target critic networks are updated through soft updates as in the standard SAC algorithm.

The feedforward SAC baseline and the recurrent SAC-LSTM architecture are illustrated in Figs. 5 and 6, respectively. Fig. 5 shows the actor and twin-critic structure of the feedforward SAC baseline. Fig. 6 shows the recurrent extension used in the proposed SAC-LSTM policy, where the previous observation–action pair is encoded by the LSTM branch and fused with the corresponding current actor or critic branch.

Compared with the feedforward SAC baseline, the SAC-LSTM architecture retains the standard actor–critic optimization structure of SAC while augmenting the instantaneous observation with a recurrent memory representation. The online recurrent-state propagation and the corresponding fixed-length recurrent training procedure are described in Section 4.2.

3.2. Recurrent sequence training

The SAC-LSTM architecture introduced in Section 4.1 requires temporally ordered samples for recurrent training. Unlike the feedforward SAC baseline, which samples individual transitions independently from the replay buffer, the recurrent policy is trained using fixed-length consecutive subsequences reconstructed from stored trajectories. This procedure preserves the off-policy training framework of SAC while allowing the LSTM branch to learn from observation–action history.

During environment interaction, the policy is executed step by step. The recurrent hidden and cell states are carried forward within each

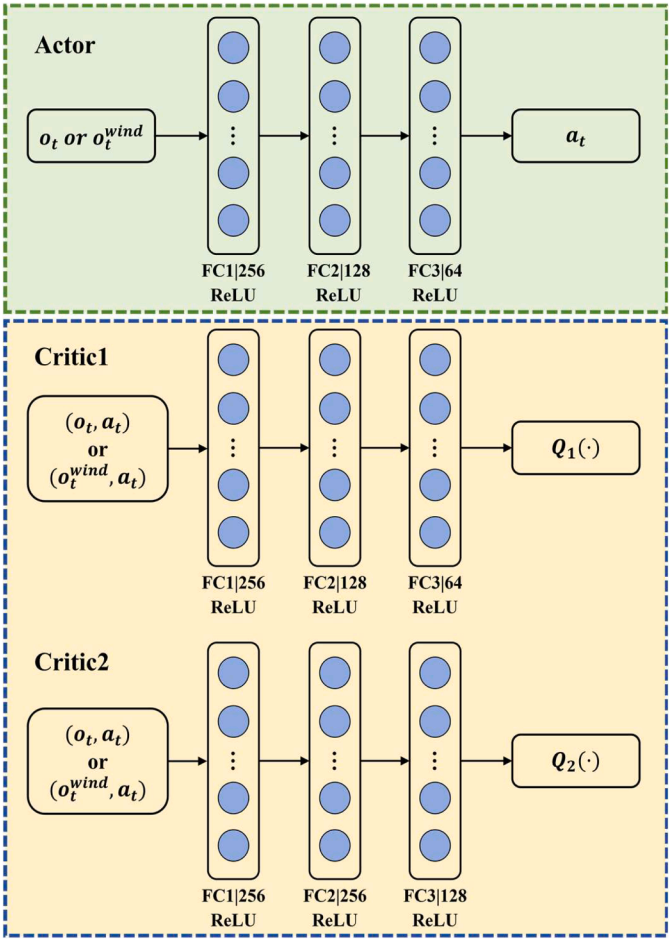


Fig. 5. Neural network structure of the SAC.

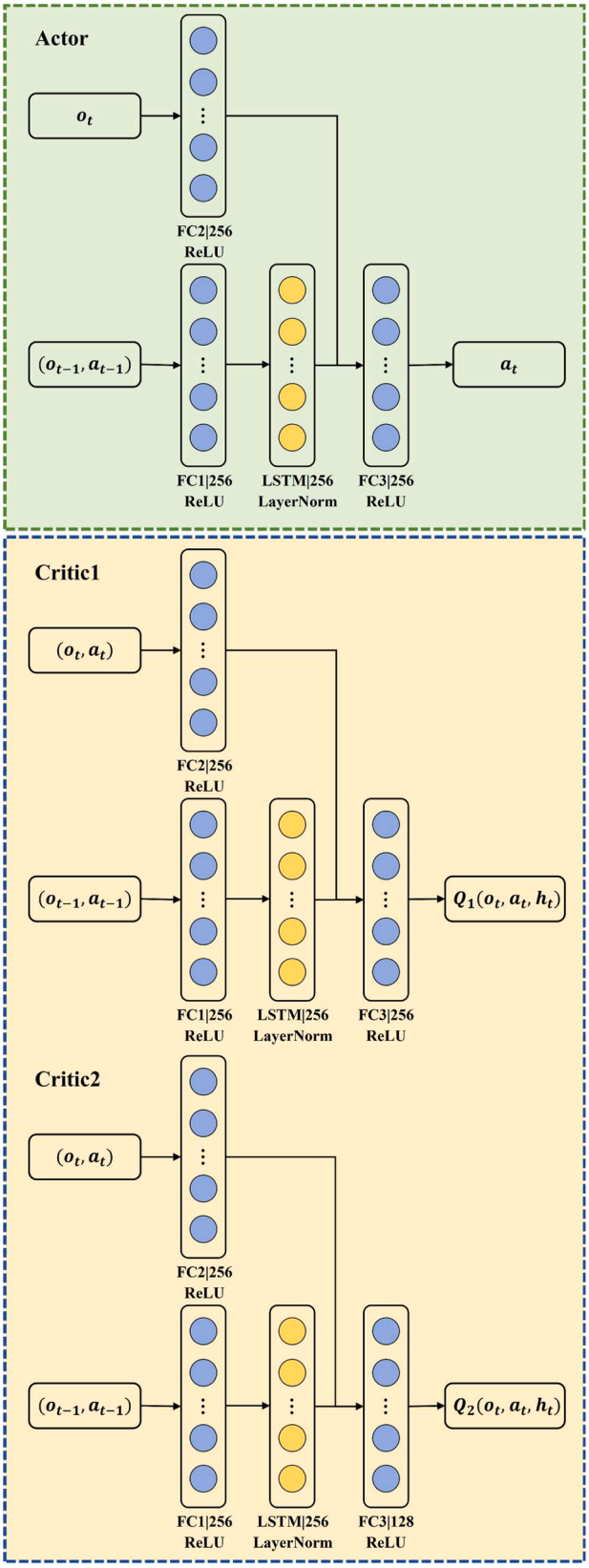


Fig. 6. Neural network structure of the SAC-LSTM.

episode and reset when a new episode begins. Each transition is stored in the replay buffer together with its episode index and temporal order. During network updates, these ordered records are used to reconstruct consecutive trajectory segments. For a sampled segment starting from time index j, the recurrent training subsequence is denoted as

$$\mathscr{B}_{j:j+K-1} = \left\{ \left(o_{t-1}, a_{t-1}, o_t, a_t, r_t, o_{t+1}, I_{\text{done},t} \right) \right\}_{t=j}^{j+K-1}, \tag{46}$$

where K is the recurrent training sequence length. In each tuple, o_{t-1} and a_{t-1} denote the previous observation–action pair used by the LSTM branch; o_t denotes the current policy observation; a_t is the executed sail–rudder action; r_t is the reward; o_{t+1} is the next observation; and $I_{\text{done},t}$ is the terminal indicator.

During recurrent unrolling, each sampled subsequence is processed in temporal order. At each time step t, the previous observation–action pair (o_{t-1}, a_{t-1}) updates the recurrent memory according to Section 4.1. The resulting memory representation h_t is then used together with the corresponding current inputs for actor and critic evaluation. For target-value computation, the recurrent memory is advanced using the current observation–action pair (o_t, a_t), so that the target critics evaluate the next observation–action pair under the corresponding next memory state.

For each network update, a minibatch of fixed-length subsequences is sampled from the replay buffer. The recurrent networks are unrolled over K consecutive control steps, and the actor and critic losses defined in Section 4.1 are evaluated over the valid time steps of the sampled subsequences. In this implementation, K is set to 10, as reported in Table II. This value denotes the fixed unroll length used for truncated recurrent training, whereas online evaluation relies on recursive propagation of the LSTM hidden and cell states at each control step. Therefore, the policy maintains temporal context through recurrent-state propagation and does not require a fixed-length history window to be reconstructed during inference.

At the beginning of each episode, the LSTM hidden and cell states are initialized to zero. During online interaction and evaluation, these recurrent states are propagated recursively throughout the episode and reset only when a new episode begins. During training, each fixed-length subsequence is sampled independently from the replay buffer, and the hidden and cell states are initialized to zero at the beginning of every sampled subsequence. When a subsequence starts from the middle of an episode, the stored preceding observation–action pair is used as its first recurrent input, but no recurrent state from before the sampled subsequence is stored or reconstructed. The network is then unrolled over the Kconsecutive steps of the sampled subsequence. The recurrent SAC-LSTM training process is summarized in Algorithm 1.

3.3. Reward function

The reward function is designed to guide the learning agent toward the target while maintaining feasible and smooth sail–rudder control behavior. The observation vector o_t has been defined in Section 3.3; this subsection focuses on the reward-related quantities and the construction of the step reward. The reward consists of dense progress feedback, an action-variation penalty, sparse checkpoint and arrival rewards, and a constant time-step penalty. For the wind-informed SAC reference, an additional wind-direction-related penalty is used because this baseline explicitly receives wind information. In the reinforcement learning formulation, the subscript t denotes the discrete control step. Variables with subscript $t$are evaluated at the current sampling instant, whereas $\left(x_f, y_f\right)$ denotes the fixed target position of the current point-to-point navigation task.

Let $\left(x_0, y_0\right)$ denote the initial position of the current navigation task. The Euclidean distance between the vessel and the target at time step t is defined as

$$L_t = \sqrt{\left(x_f - x_t\right)^2 + \left(y_f - y_t\right)^2} \tag{47}$$

The distance-reduction term is then calculated as

Algorithm 1
Recurrent SAC training for the wind-input-free SAC-LSTM policy.

Input: recurrent training sequence length K, replay buffer capacity N, minibatch size B, discount factor γ, temperature coefficient α, soft update coefficient τ, actor learning rate η_π, and critic learning rate η_Q.
Initialize: actor network π_θ with an LSTM module; twin critic networks Q_{ω_1} and Q_{ω_2}; target critic networks $Q_{\bar{\omega}_1} \leftarrow Q_{\omega_1}$ and $Q_{\bar{\omega}_2} \leftarrow Q_{\omega_2}$; replay buffer $\mathscr{D}$.
1. For each episode do
2. Reset the simulation environment and obtain the initial observation o_0.
3. Initialize the LSTM hidden and cell states to zero.
4. For each control step t do
5. If $t = 0$, use the zero-initialized recurrent memory for action selection.
6. If $t \geq 1$, construct the recurrent input $z_t = [o_{t-1}; a_{t-1}]$ and update the recurrent states.
7. Generate the sail–rudder action a_t using the actor with the current observation o_t and recurrent memory h_t.
8. Execute a_t in the simulation environment.
9. Receive r_t, o_{t+1}, and $I_{\text{done},t}$.
10. Store the ordered transition
$(o_{t-1}, a_{t-1}, o_t, a_t, r_t, o_{t+1}, I_{\text{done},t})$
in the replay buffer $\mathscr{D}$, while preserving its episode index and temporal order.
11. If $I_{\text{done},t} = 1$, terminate the current episode.
12. End For
13. For each network update do
14. Sample B fixed-length consecutive subsequences from $\mathscr{D}$, each with length K:
$\mathscr{B}_{jj+K-1} = \{(o_{t-1}, a_{t-1}, o_t, a_t, r_t, o_{t+1}, I_{\text{done},t})\}_{t=j}^{j+K-1}$.
15. Initialize the LSTM hidden and cell states to zero at the beginning of each sampled subsequence.
Unroll the recurrent networks over K time steps.
17. Compute the SAC critic loss over the sampled subsequences using the formulation in Section 4.1.
18. Update the critic networks Q_{ω_1} and Q_{ω_2}.
19. Compute the SAC actor loss over the sampled subsequences using the formulation in Section 4.1.
20. Update the actor network π_θ.
21. Update the temperature coefficient α, if adaptive entropy tuning is used.
22. Soft-update the target critic networks:
$\bar{\omega}_i \leftarrow \tau\omega_i + (1-\tau)\bar{\omega}_i, i = 1, 2$.
23. End For.
24. End For.

$$\Delta L_t = L_{t-1} - L_t. \tag{48}$$

A positive ΔL_t indicates that the vessel moves closer to the target between two consecutive sampling steps, whereas a negative value indicates temporary motion away from the target.

However, distance reduction alone may provide insufficient or even misleading feedback under headwind conditions. In such cases, the vessel may need to perform tacking maneuvers, during which short-term motion does not necessarily lead to a monotonic reduction in the direct distance to the target. Therefore, an additional projection-based progress term is introduced. The straight-line distance from the initial position to the target is

$$D_o = \sqrt{(x_f - x_o)^2 + \left(y_f - y_o\right)^2} \tag{49}$$

The projected progress of the vessel along the start-to-target direction is defined as

$$l_t = \frac{\left(x_t - x_0\right)\left(x_f - x_0\right) + \left(y_t - y_0\right)\left(y_f - y_0\right)}{D_0}. \tag{50}$$

Accordingly, the incremental projected progress is calculated as

$$\Delta l_t = l_t - l_{t-1}. \tag{51}$$

Compared with ΔL_t, which directly measures the reduction in target distance, Δl_t provides complementary directional feedback along the nominal start-to-target direction. This design helps maintain a useful learning signal when the vessel follows curved or zigzag trajectories, particularly under headwind conditions. The geometric interpretation of ΔL_t and Δl_t is illustrated in Fig. 7.

Building upon the distance-reduction and projected-progress definitions above, the reward function for each control step is formulated as.

$$R_t = \lambda_t\left(R_{progress,t} + R_{action,t}\right) + R_{checkpoint,t} + R_{arrival,t} + R_{penalty,t}. \tag{52}$$

The coefficient λ_t adaptively scales the dense progress and action-variation terms:

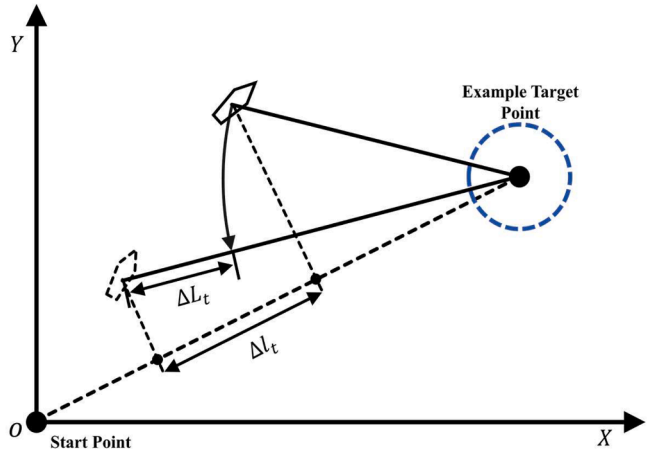


Fig. 7. The representation of differential state variables.

$$\lambda_t = \lambda_0 + \lambda_{s,t} + \lambda_{r,t}, \lambda_0 = 0.2. \tag{53}$$

The additional terms $\lambda_{s,t}$ and $\lambda_{r,t}$ increase this coefficient slightly when the sail and rudder adjustments remain sufficiently small:

$$\lambda_{s,t} = \begin{cases} 0.1\lambda_0, & \left|\Delta\delta_{s,t}\right| \leq \delta, \\ 0, & \text{otherwise}, \end{cases} \lambda_{r,t} = \begin{cases} 0.1\lambda_0, & \left|\Delta\delta_{r,t}\right| \leq \delta, \\ 0, & \text{otherwise}, \end{cases} \delta = \frac{\pi}{180}. \tag{54}$$

The progress reward combines target-distance reduction, projected directional progress, and a surge-velocity term moderated by lateral velocity magnitude:

$$R_{\text{progress},t} = 2\Delta L_t + \Delta l_t + \frac{0.5u_t}{1 + |v_t|}. \tag{55}$$

The action penalty discourages large incremental sail and rudder adjustments:

$$R_{\text{action},t} = -\left\lfloor \frac{|\Delta\delta_{s,t}|}{\delta} \right\rfloor \cdot 0.5 - \left\lfloor \frac{|\Delta\delta_{r,t}|}{\delta} \right\rfloor \cdot 0.5, \delta = \frac{\pi}{180}. \tag{56}$$

The checkpoint reward provides sparse milestone-based shaping and is awarded only when the vessel reaches each distance threshold for the first time:

$$R_{\text{checkpoint},t} = 3\mathbb{I}_{\text{first}}(L_t \le 375) + 5\mathbb{I}_{\text{first}}(L_t \le 250) + 12\mathbb{I}_{\text{first}}(L_t \le 125), \tag{57}$$

where $\mathbb{I}_{\text{first}}(\cdot)$ equals 1 only at the first time step when the corresponding condition is satisfied, and equals 0 otherwise.

The terminal arrival reward is defined as.

$$R_{\text{arrival},t} = \begin{cases} 40, & L_t \le 10, \\ 0, & \text{otherwise}. \end{cases} \tag{58}$$

A constant time-step penalty is further introduced at each control step to encourage efficient task completion:

$$R_{\text{penalty},t} = -0.2. \tag{59}$$

For the SAC-wind baseline, the policy receives the augmented wind-input observation defined in Section 3.3. Since this baseline explicitly receives wind-direction information, a no-go-zone penalty is introduced as physically motivated prior guidance during learning. This term discourages heading choices that are inconsistent with basic sailing constraints and helps regulate early-stage exploration in the wind-aware setting. Accordingly, SAC-wind serves as a wind-informed learning reference combining explicit wind input with wind-direction-based guidance. The effect of recurrent history is assessed through the comparison between SAC-nowind and SAC-LSTM, which share the same wind-input-free current observation. The wind-direction difference is first computed as

$$\Delta\psi_{w,t} = \text{wrap}_{[-\pi,\pi]}\left(\psi_t \widehat{\psi}_{w,t}\right). \tag{60}$$

The reward function for the wind-informed SAC reference is then written as.

$$R_t^{\text{wind}} = \lambda_t\left(R_{\text{progress},t} + R_{\text{action},t}\right) + R_{\text{checkpoint},t} + R_{\text{arrival},t} + R_{\text{penalty},t} + R_{\text{no-go},t}, \tag{61}$$

where.

$$R_{\text{no-go},t} = \begin{cases} -2, & |\Delta\psi_{w,t}| > \theta_{\text{no-go}}, \\ 0, & \text{otherwise}, \end{cases} \theta_{\text{no-go}} = \frac{3\pi}{4}. \tag{62}$$

Simulation Results and Analysis.

This section evaluates the proposed SAC-LSTM policy under the wind-input-free partially observable navigation setting defined in Section 3. The main purpose is to examine whether recurrent observation–action history improves point-to-point sail–rudder control when measured wind speed and wind direction are unavailable to the policy. Three learning-based policies are compared: SAC-nowind, SAC-wind, and SAC-LSTM. In addition, MPC is included as a model-based reference under representative wind directions because it uses explicit model knowledge and wind information and requires higher online computational cost.

3.4. Experimental setup

The simulation experiments were conducted under the point-to-point navigation task formulated in Section 3. The environment was based on the 4-DOF sailing vessel dynamics described in Section 3.1, together with the Dryden wind-field model and observation-noise setting introduced in Section 3.2. This subsection summarizes the task configuration, wind conditions, compared methods, and training settings. The evaluation protocol and statistical reporting are described in Section 5.1.1.

The navigation task was defined from the initial position (0, 0) to the target position (500, 0). An evaluation episode was considered successful if the vessel reached a circular target region with a radius of 10 m within the maximum episode length. The maximum episode length was set to 1000 control steps. The control and wind-field update frequencies were both set to 1 Hz, and the vessel dynamics were integrated using the Runge–Kutta scheme with a base integration step of 0.05 s. The sail and rudder angles were constrained to [-90°, 90°] and [-30°, 30°], respectively, and the incremental sail and rudder commands were limited to [-10°, 10°] per control step.

In the randomized-wind evaluation, the mean wind speed was set to 8 m/s, and the initial mean wind direction was randomly sampled from [-180°, 180°] at the beginning of each episode. In the representative-wind evaluation, the initial mean wind direction was set to one of four representative values covering typical sailing conditions: 0° tailwind, 90° crosswind, 150° oblique headwind, and 180° headwind. Under this convention, 0° denotes the case in which the initial mean wind direction is aligned with the route from the start point to the target, while 180° denotes the corresponding headwind condition.

For each learning-based method, five policies were trained independently using the training configurations and random seeds listed in Table II, and all five trained policies were included in the evaluation. In the randomized-wind evaluation, each trained policy was evaluated over 500 episodes. In the representative-wind evaluation, each trained policy was evaluated over 125 episodes for each representative initial mean wind direction. The construction and matching of the evaluation scenarios are described in Section 5.1.1.

Table I summarizes the main simulation settings and evaluation cases used in this study.

The three learning-based policies differ in their access to wind information and recurrent memory. SAC-nowind serves as a memoryless wind-input-free SAC baseline that uses the current onboard observable vessel states and target-relative information without direct wind measurements. SAC-wind serves as a wind-informed learning reference that augments the observation vector with noisy wind speed and wind direction measurements and uses the wind-direction-based no-go penalty defined in Section 4.3. SAC-LSTM is the proposed recurrent wind-input-

Table I
Simulation and evaluation settings.

Category	Setting
Navigation task	Start point: (0, 0); target point: (500, 0); success radius: 10 m; maximum episode length: 1000 control steps
Wind condition	Dryden wind field as defined in Section 3.2; mean wind speed: 8 m/s; initial mean wind direction randomly sampled from [-180°, 180°], unless a representative mean wind direction is specified
Representative initial mean wind directions	0° tailwind, 90° crosswind, 150° oblique headwind, and 180° headwind
Observation noise	Nominal onboard observation noise as defined in Section 3.2; noisy wind measurements are provided only to SAC-wind
Actuator constraints	Sail angle: [-90°, 90°]; rudder angle: [-30°, 30°]; incremental sail/rudder commands: [-10°, 10°] per control step
Simulation frequency	Control frequency: 1 Hz; wind-field update frequency: 1 Hz; dynamics integration base step: 0.05 s
Learning-based policies	SAC-nowind, SAC-wind, and SAC-LSTM; five independent training policies for each method
Evaluation episodes	Randomized wind: 500 episodes per trained policy; representative wind: 125 episodes per trained policy and representative initial mean wind direction
Reported metrics	Task success rate, average time consumption, average trajectory length, sail-angle variation rate, and rudder-angle variation rate
MPC reference	Evaluated under representative wind directions and reported separately as a model-based reference because it uses explicit model and wind information and has a different evaluation sample size

free policy. It uses the same wind-input-free current observation as SAC-nowind, while incorporating recent observation–action history through the recurrent architecture described in Section 4.1. The comparison between SAC-nowind and SAC-LSTM examines the role of recurrent observation–action history under the same wind-input-free current-observation setting.

A model predictive control (MPC) method was also included as a model-based reference for the representative-wind evaluation. The MPC controller uses the same sailing vessel dynamics model and simulation environment, together with the current state, target position, and explicit wind information, to solve an online finite-horizon optimization problem. The prediction horizon was set to 20 steps, and the control horizon was set to 1 step. MPC therefore uses explicit model and wind information and performs online optimization at each control step. Its results are reported separately as a model-based reference for the representative-wind evaluation.

Table II lists the training hyperparameters of the learning-based policies. SAC-nowind and SAC-wind share the same feedforward SAC configuration. SAC-LSTM uses the same SAC optimization settings with an additional recurrent branch. The corresponding network architectures have been described in Section 4.1 and illustrated in Figs. 5 and 6. The recurrent training sequence length was fixed at $K = 10$ for the main experiments and denotes the unroll length used during recurrent training, as described in Section 4.2. During online evaluation, temporal information was maintained through recursive propagation of the LSTM hidden and cell states.

3.4.1. Evaluation protocol and statistical reporting

To account for variability across independent training runs and stochastic evaluation conditions, the learning-based policies were evaluated in five matched evaluation blocks. Each block contained one independently trained policy from each compared method or reward configuration and a shared bank of episode-level evaluation cases. Within each block, corresponding episodes with the same case identifier were initialized using the same initial mean wind direction, vessel heading and velocity, and target position. They were also driven by the same pre-generated Dryden wind-disturbance sequence and the same observation-noise sequence. Because the policies generated different actions, their resulting vessel trajectories, state histories, and apparent-wind histories were policy-specific. When corresponding episodes terminated at different times, the wind and observation-noise sequences were identical over their common elapsed prefix. The five blocks used independently generated evaluation scenario banks. These shared initial conditions and exogenous stochastic inputs established the pairing within each matched block, while the training-seed label only identified the independently trained policy assigned to that block.

Each independently trained policy was treated as one replicate. Episode-level outcomes were summarized within each policy to obtain one seed-level estimate for each metric, and episode-level observations were not treated as independent replicates. Task success rate was calculated over all evaluation episodes, while average time consumption, average trajectory length, sail-angle variation rate, and rudder-angle variation rate were calculated over successful episodes. Final distance and total reward in the reward-component ablation were calculated over all evaluation episodes. Tables III, IV, and VI report the mean ± between-seed standard deviation and two-sided 95% confidence interval across the five seed-level estimates. The confidence intervals were calculated using Student's t distribution with four degrees of freedom. For task success rate, interval endpoints outside the admissible range were truncated to 0% or 100% for display.

Paired comparisons were constructed from within-block differences between corresponding seed-level estimates and are reported in the Supplementary Material. Each difference was calculated as the first-listed method or case minus the second-listed method or case. The supplementary comparisons report nonparametric percentile bootstrap confidence intervals, exact two-sided sign-flip p-values, Holm-adjusted p-values, and paired Cohen's d_z. For the randomized-wind evaluation, each metric defined a separate Holm family containing the comparisons of SAC-LSTM with SAC-nowind and SAC-wind. For the representative-wind evaluation, each initial mean wind direction × metric combination defined a separate family containing only the estimable focal comparisons. For the reward-component analysis, each metric defined a family containing the four comparisons between Full SAC-LSTM and the ablation configurations.

With five paired observations, the exact sign-flip test enumerates $2^5 = 32$ sign assignments, and the smallest attainable two-sided p-value is 0.0625. None of the exact paired tests reported in the Supplementary Material reached 0.05. Bootstrap intervals based on five paired differences may be unstable or appear overly narrow, and very large paired Cohen's d_z values can arise when the variability among the paired differences is very small. These quantities are therefore interpreted as supplementary descriptive evidence, and d_z is not classified using conventional small, medium, or large effect-size thresholds.

3.5. Training performance

The training behavior of the three learning-based policies was first examined under the same total number of training steps and optimization settings. Fig. 8 shows the evaluation return, critic loss, and actor loss of SAC-nowind, SAC-wind, and SAC-LSTM over five independent training seeds. These curves are used as training diagnostics for examining learning trends and optimization stability, whereas final navigation performance is evaluated in the dedicated test scenarios.

As shown in Fig. 8(a), the evaluation returns of all three policies

Table II
Training hyperparameters of the learning-based policies.

Parameter	SAC-nowind / SAC-wind	SAC-LSTM
Discount factor	0.98	0.98
Actor learning rate	3×10^{-4}	3×10^{-4}
Critic learning rate	3×10^{-4}	3×10^{-4}
Temperature coefficient	0.2	0.2
Temperature learning rate	3×10^{-5}	3×10^{-5}
Replay buffer capacity	1×10^{6}	1×10^{6}
Batch size	256	256
Soft update coefficient	0.002	0.002
Maximum steps per episode	1000	1000
Recurrent training sequence length (K)	—	10
Total training steps	7.5×10^{6}	7.5×10^{6}
Independent training seeds	0,100,200,300,400	0,100,200,300,400

Table III
Randomized-wind evaluation of learning-based policies.

Metric	SAC-nowind	SAC-wind	SAC-LSTM
Task success rate(%)	39.64 ± 1.65 [37.59, 41.69]	95.24 ± 0.38 [94.76, 95.72]	99.36 ± 0.26 [99.04, 99.68]
Average time consumption (s)	207.64 ± 6.98 [198.98, 216.30]	178.77 ± 1.92 [176.38, 181.16]	177.22 ± 1.61 [175.23, 179.22]
Average trajectory length (m)	610.25 ± 2.93 [606.62, 613.89]	584.81 ± 2.49 [581.72, 587.89]	586.38 ± 4.15 [581.24, 591.53]
Sail angle variation rate (°/s)	1.195 ± 0.022 [1.168, 1.223]	0.865 ± 0.014 [0.847, 0.883]	0.875 ± 0.008 [0.866, 0.885]
Rudder angle variation rate (°/s)	0.840 ± 0.017 [0.819, 0.861]	0.675 ± 0.009 [0.664, 0.687]	0.633 ± 0.013 [0.617, 0.649]

Remark: Values are reported as mean ± between-seed SD [95% CI] across five seed-level estimates from independently trained policies. The 95% CIs are two-sided Student's t intervals with four degrees of freedom. For task success rate, interval endpoints outside [0%, 100%] are truncated to the admissible range for display. Metric-specific episode inclusion follows Section 5.1.1.

Table IV
Representative-wind evaluation of learning-based policies.

Wind Direction	Evaluation metrics	SAC-nowind	SAC-wind	SAC-LSTM
	Task success rate(%)	90.56 ± 1.91 [88.19, 92.93]	100.00 ± 0.00 [100.00, 100.00]	99.84 ± 0.36 [99.40, 100.00]
	Average time consumption(s)	196.08 ± 6.80 [187.64, 204.52]	148.98 ± 1.23 [147.46, 150.51]	148.46 ± 1.10 [147.10, 149.83]
0°	Average trajectory length(m)	607.93 ± 7.61 [598.49, 617.38]	516.00 ± 1.53 [514.11, 517.90]	509.99 ± 1.14 [508.57, 511.41]
	Sail angle variation rate(°/s)	1.086 ± 0.027 [1.053, 1.120]	0.657 ± 0.011 [0.644, 0.671]	0.584 ± 0.018 [0.561, 0.606]
	Rudder angle variation rate(°/s)	0.839 ± 0.011 [0.826, 0.853]	0.480 ± 0.027 [0.446, 0.514]	0.470 ± 0.014 [0.453, 0.487]
	Task success rate(%)	66.24 ± 2.43 [63.23, 69.25]	100.00 ± 0.00 [100.00, 100.00]	99.68 ± 0.44 [99.14, 100.00]
	Average time consumption(s)	227.40 ± 19.77 [202.85, 251.94]	145.91 ± 1.70 [143.79, 148.02]	153.61 ± 2.80 [150.14, 157.08]
90°	Average trajectory length(m)	639.13 ± 18.19 [616.55, 661.72]	531.35 ± 3.40 [527.12, 535.58]	548.56 ± 2.06 [546.01, 551.12]
	Sail angle variation rate(°/s)	1.375 ± 0.019 [1.351, 1.398]	0.870 ± 0.023 [0.842, 0.898]	1.094 ± 0.014 [1.076, 1.111]
	Rudder angle variation rate(°/s)	0.792 ± 0.009 [0.780, 0.804]	0.573 ± 0.014 [0.555, 0.591]	0.606 ± 0.021 [0.580, 0.632]
	Task success rate(%)	0.00 ± 0.00 [0.00, 0.00]	78.24 ± 3.17 [74.30, 82.18]	99.20 ± 0.57 [98.50, 99.90]
	Average time consumption(s)	NA	281.58 ± 6.62 [273.37, 289.80]	238.94 ± 2.39 [235.98, 241.91]
150°	Average trajectory length(m)	NA	736.56 ± 8.52 [725.99, 747.14]	733.97 ± 3.59 [729.52, 738.43]
	Sail angle variation rate(°/s)	NA	0.758 ± 0.018 [0.736, 0.780]	1.161 ± 0.021 [1.135, 1.186]
	Rudder angle variation rate(°/s)	NA	0.852 ± 0.012 [0.838, 0.867]	0.762 ± 0.010 [0.750, 0.775]
	Task success rate(%)	0.00 ± 0.00 [0.00, 0.00]	88.80 ± 4.42 [83.31, 94.29]	99.20 ± 1.39 [97.48, 100.00]
	Average time consumption(s)	NA	299.68 ± 4.30 [294.34, 305.02]	293.24 ± 5.82 [286.01, 300.47]
180°	Average trajectory length(m)	NA	793.11 ± 3.41 [788.88, 797.35]	812.82 ± 5.86 [805.55, 820.09]
	Sail angle variation rate(°/s)	NA	0.965 ± 0.013 [0.949, 0.981]	1.082 ± 0.020 [1.057, 1.107]
	Rudder angle variation rate(°/s)	NA	1.055 ± 0.011 [1.042, 1.069]	1.073 ± 0.009 [1.061, 1.085]

Remark: Values are reported as mean ± between-seed SD [95% CI] across five seed-level estimates from independently trained policies. The 95% CIs are two-sided Student's t intervals with four degrees of freedom. For task success rate, interval endpoints outside [0%, 100%] are truncated to the admissible range for display. Metric-specific episode inclusion follows Section 5.1.1.

increase during the early training stage, indicating that the agents begin to acquire task-relevant sail–rudder control behaviors through interaction with the simulation environment. After this initial improvement, however, the learning profiles differ across policies. SAC-nowind, which relies only on the current wind-input-free observation, converges to a lower return level across all five training seeds, indicating greater difficulty in learning reliable sail–rudder behavior under hidden wind conditions. Without direct wind input or recurrent temporal context, the current onboard observation and target-relative information may be insufficient for consistent decision-making across varying wind scenarios.

SAC-wind and SAC-LSTM both reach higher evaluation-return levels than SAC-nowind. For SAC-wind, the additional wind measurements and wind-direction-based guidance provide direct wind-related information during learning. For SAC-LSTM, the improvement is achieved without measured wind speed or wind direction as policy inputs. Instead, the recurrent branch allows the policy to use recent observation–action history as temporal context for action selection. In Fig. 8(a), SAC-LSTM reaches a high evaluation-return region earlier than SAC-wind, although SAC-wind gradually approaches a similar return level toward the end of training. The dispersion among the training-seed trends of SAC-wind and SAC-LSTM is broadly comparable, while the training curves show different convergence profiles and achieved return levels.

The critic and actor losses in Fig. 8(b) and Fig. 8(c) are used to inspect the optimization process rather than to rank the final policies directly. The loss curves do not show obvious divergence, suggesting that the actor–critic updates remain numerically stable under the selected hyperparameters. However, the magnitude of the actor or critic loss alone is not a reliable measure of navigation performance in reinforcement learning. Therefore, the evaluation return is used here to characterize training progress, while the final navigation capability is assessed through the randomized-wind and representative-wind evaluations in the following sections.

Overall, the training diagnostics show distinct learning profiles across the three policies. SAC-LSTM reaches higher evaluation-return levels than SAC-nowind without direct wind input, while SAC-wind approaches a similar return region toward the end of training. Final navigation performance is assessed in Section 5.3, and the corresponding seed-level results are provided in the Supplementary Material.

3.6. Comparative performance under wind conditions

After training, dedicated test evaluations were conducted to assess the final navigation performance of the learned policies. The randomized-wind evaluation served as the main benchmark for comparing SAC-nowind, SAC-wind, and SAC-LSTM across randomized hidden wind conditions. Because SAC-nowind and SAC-LSTM use the

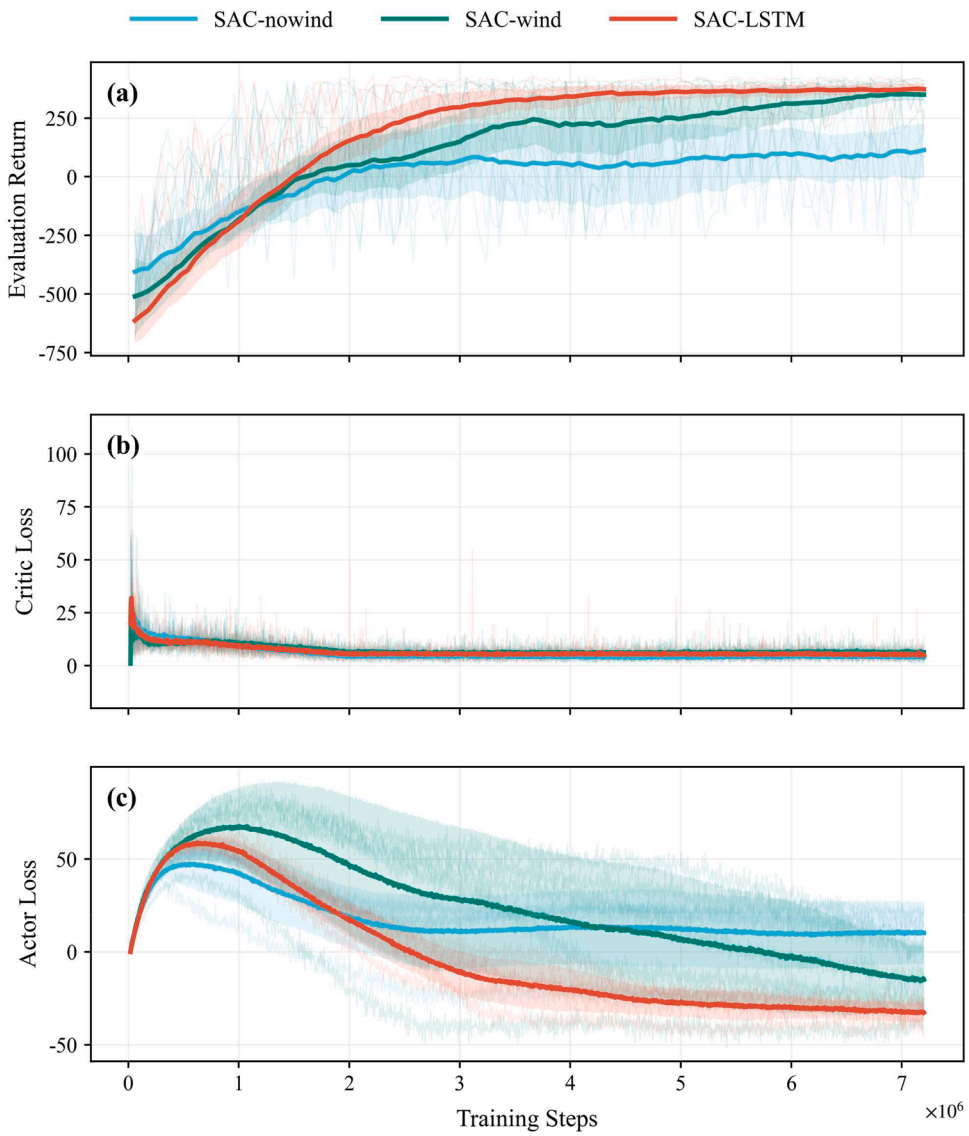


Fig. 8. Training diagnostics of the learning-based policies across five independent training seeds. (a) Evaluation return; (b) critic loss; and (c) actor loss.
Remark: The bold curves denote the mean smoothed trends, the shaded regions denote the standard deviation across the five seeds, and the light curves show the recorded values before smoothing.

same wind-input-free current observation, their comparison focuses on the role of recurrent observation–action history under partial observability. SAC-wind is included as a wind-informed learning reference, as it receives noisy wind measurements and uses the wind-direction-related guidance term defined in Section 4.3.

The randomized-wind evaluation results are summarized in Table III using the statistical reporting protocol described in Section 5.1.1.

The randomized-wind benchmark captures the main challenge of the proposed wind-input-free setting: the policy must select sail–rudder actions under continuously varying wind conditions that are hidden from the policy. As shown in Table III, SAC-nowind achieved a task success rate of 39.64 ± 1.65%, whereas SAC-LSTM achieved 99.36 ± 0.26% under the same wind-input-free current-observation setting. This comparison suggests that the current observation alone was not sufficient for reliable control across the randomized wind conditions, whereas recurrent observation–action history provided useful temporal context for sail–rudder decision-making under partial observability.

The successful-episode metrics further characterize navigation efficiency and control behavior among completed trials. Compared with SAC-nowind, SAC-LSTM reduced the average time consumption from 207.64 ± 6.98 s to 177.22 ± 1.61 s. It also reduced the rudder-angle variation rate from 0.840 ± 0.017°/s to 0.633 ± 0.013°/s and the sail-angle variation rate from 1.195 ± 0.022°/s to 0.875 ± 0.008°/s. These results suggest that the recurrent policy not only completed more randomized-wind trials but also produced smoother sail–rudder commands among successful episodes.

Compared with the wind-informed SAC reference, SAC-LSTM achieved a higher task success rate without using measured wind speed or wind direction as policy inputs. Its average time consumption was close to that of SAC-wind, and the two methods showed similar trajectory length and sail-angle variation rate. SAC-LSTM also produced a lower rudder-angle variation rate. Under the randomized-wind setting, the main differences between the two policies were therefore observed in task completion and rudder-command variation, while the remaining successful-episode metrics were broadly comparable.

The comparison between SAC-LSTM and SAC-wind is further examined across the five matched evaluation blocks because the two policies show relatively close values in several successful-episode metrics. Fig. 9 therefore complements Table III by showing the performance of the independently trained policies under randomized wind

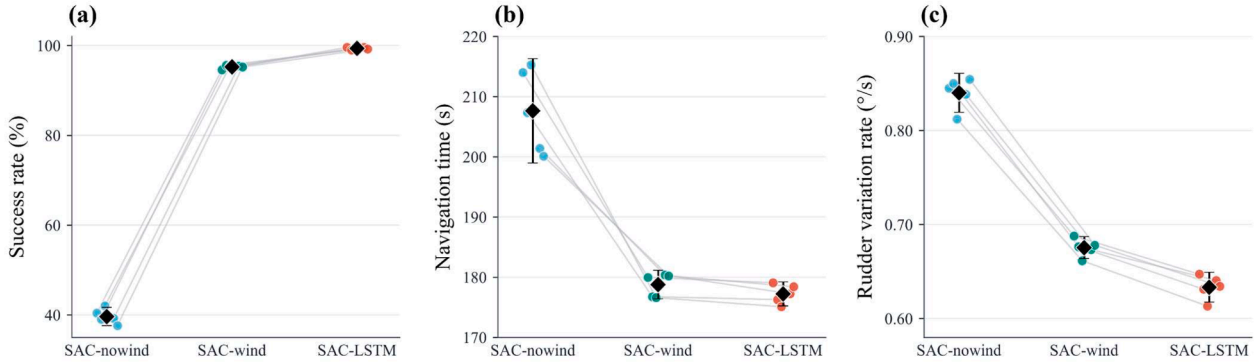


Fig. 9. Seed-level comparison of learning-based policies under randomized wind conditions.

conditions. Each marker corresponds to one trained policy evaluated using the scenario bank of its matched block. This visualization shows the consistency of the main patterns in Table III across the five training seeds.

Fig. 9 provides a seed-level view of the randomized-wind results reported in Table III. Each point represents the mean performance of one independently trained policy over its randomized-wind evaluation episodes. Across the five seeds, SAC-nowind consistently shows lower task success rates than SAC-wind and SAC-LSTM. SAC-LSTM maintains the highest completion rate and lower rudder-angle variation, while its successful-episode average time consumption remains close to that of SAC-wind. This pattern is consistent with the numerical summary in Table III and suggests that the main differences among the learning-based policies are not driven by a single trained run.

The randomized-wind benchmark evaluates policy performance over a broad range of wind directions, but it does not show how the policies behave under specific sailing conditions. To make the wind-direction-dependent behavior more interpretable, four representative wind directions were further evaluated: 0° tailwind, 90° crosswind, 150° oblique headwind, and 180° headwind. These cases correspond to typical sailing regimes with different requirements for sail–rudder coordination. The learning-based results are summarized in Table IV.

Table IV further separates the learning-based results into four representative wind directions. This analysis examines how the policies behave under typical sailing conditions, rather than replacing the randomized-wind benchmark. The primary wind-input-free comparison is between SAC-nowind and SAC-LSTM, because both policies use the same current observation and differ mainly in whether recent observation–action history is encoded. Across all four wind directions, SAC-LSTM achieves higher task success rates than SAC-nowind. SAC-nowind remains feasible in the tailwind and crosswind cases, but no successful episodes are observed at 150° or 180°. This result suggests that the current wind-input-free observation alone does not provide sufficient information for reliable sail–rudder coordination in the more demanding headwind-related cases.

SAC-wind achieved 100% task success rates in the 0° and 90° cases and remained feasible at 150° and 180°, although its success rates in the two headwind-related cases were lower than those of SAC-LSTM under the present training and reward settings. These results characterize the wind-direction-dependent performance of the feedforward wind-informed policy used in this study.

The continuous metrics are reported together with task success because time consumption, trajectory length, and control-variation rates are computed only from successful episodes. The NA entries for SAC-nowind at 150° and 180° therefore reflect the absence of successful episodes rather than a separate continuous-metric comparison. SAC-LSTM does not achieve the lowest value for every continuous metric under every wind direction. For example, SAC-wind produces shorter successful-episode trajectories or lower sail-angle variation in some cases, whereas SAC-LSTM mainly maintains higher task completion under the wind-input-free setting. Across the four representative directions, recurrent observation–action history improved task completion relative to SAC-nowind, particularly at 150° and 180°. SAC-wind showed different trade-offs in the successful-episode efficiency and control-variation metrics.

To provide a compact visual summary, Fig. 10 shows the mean task success rate and sail- and rudder-angle variation rates across the five training seeds for the four representative initial mean wind directions.

Fig. 10 highlights that the largest completion-rate differences occur under the 150° and 180° initial mean wind directions. The control-variation panels should be interpreted together with task success, because low sail- or rudder-angle variation is not necessarily favorable when the corresponding completion rate is low or zero.

MPC was additionally evaluated as a model-based reference under the same representative wind directions. It uses explicit wind information and the vessel model to perform online finite-horizon optimization, and its results are reported separately in Table V.

Table V reports the MPC results under the four representative wind directions. MPC completed all 20 trials in each direction, showing that the selected representative-wind tasks are feasible for a model-based controller when the vessel model and explicit wind information are available during online optimization. The average time consumption and trajectory length vary substantially across wind directions, with longer completion times in the 150° oblique-headwind and 180° headwind cases. This pattern is consistent with the increased maneuvering demand under headwind-related sailing conditions.

Because MPC differs from the learning-based policies in information access, control mechanism, and evaluation sample size, Table V is used only for contextual comparison and to support the trajectory-level examples in Fig. 11. Fig. 11 shows selected trajectories under the four representative initial mean wind directions for SAC-nowind, SAC-wind, SAC-LSTM, and MPC, allowing successful navigation behavior and selected non-completion cases to be examined qualitatively.

Fig. 11 provides selected trajectory examples under the four representative wind directions. These trajectories are not used as statistical summaries; rather, they illustrate the navigation patterns behind the quantitative results in Tables IV and V. In the 0° tailwind and 90° crosswind cases, the selected trajectories of the compared controllers reach the target, although their path shapes differ. This is consistent with the relatively less demanding sail–rudder coordination required under these wind directions.

The differences become more pronounced in the 150° oblique-headwind and 180° headwind cases. In these conditions, direct straight-line motion is no longer representative of feasible sailing behavior, and successful trajectories tend to involve zigzag maneuvers. SAC-LSTM and MPC reach the target in the selected 150° case, whereas SAC-nowind and SAC-wind do not complete the task in this particular episode. The SAC-wind trajectory at 150° illustrates one non-completion

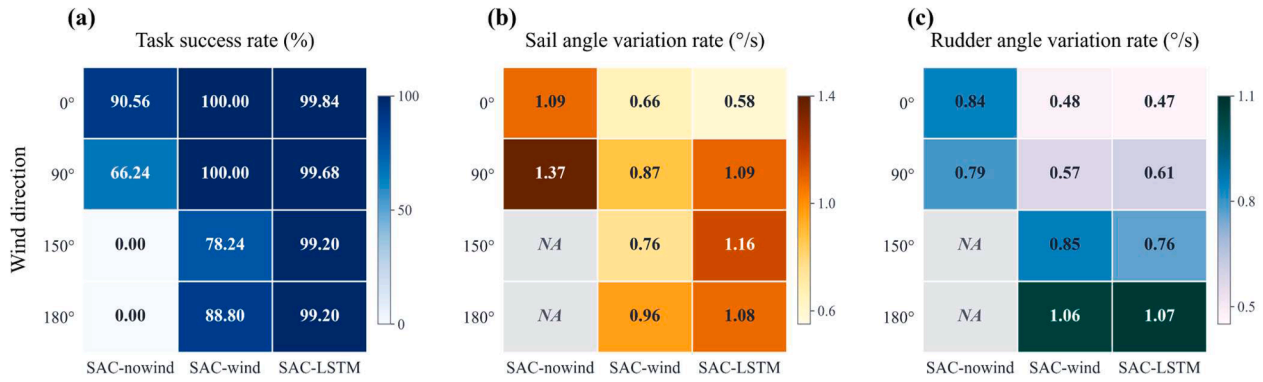


Fig. 10. Representative-wind performance of learning-based policies.

Table V
MPC Reference Under Representative Wind Directions.

Evaluation metrics	0°	90°	150°	180°
Task success rate (%)	20/20; 100 [83.89, 100]	20/20; 100 [83.89, 100]	20/20; 100 [83.89, 100]	20/20; 100 [83.89, 100]
Average time consumption(s)	190.30 ± 15.44 [183.07, 197.53]	143.05 ± 13.07 [136.93, 149.17]	310.90 ± 29.49 [297.10, 324.70]	365.44 ± 50.10 [341.99, 388.89]
Average trajectory length(m)	712.76 ± 79.78 [675.42, 750.10]	573.17 ± 42.85 [553.12, 593.22]	606.30 ± 37.03 [588.97, 623.63]	734.92 ± 49.42 [711.79, 758.05]
Sail angle variation rate(°/s)	1.08 ± 0.16 [1.01, 1.15]	0.92 ± 0.26 [0.80, 1.04]	0.42 ± 0.23 [0.31, 0.53]	0.49 ± 0.16 [0.42, 0.56]
Rudder angle variation rate (°/s)	0.22 ± 0.14 [0.15, 0.29]	0.28 ± 0.10 [0.23, 0.33]	0.44 ± 0.10 [0.39, 0.49]	0.64 ± 0.16 [0.57, 0.71]

Remark: Values are reported over 20 MPC trials for each representative wind direction. Success-rate confidence intervals are Wilson 95% intervals. Continuous-metric confidence intervals are two-sided Student's t intervals with 19 degrees of freedom. MPC is reported separately because it uses explicit model and wind information and has a different evaluation sample size from the learning-based policies.

episode and should be interpreted together with the aggregate results in Table IV.

In the displayed headwind-related cases, SAC-LSTM maintains feasible wind-input-free navigation through recurrent observation–action history, while SAC-wind and MPC exhibit trajectories under their respective wind-informed formulations.

The remaining differences in path geometry and sail–rudder behavior motivate the reward-component analysis in the next subsection.

3.7. Reward-component ablation

The preceding evaluations show that SAC-LSTM improves the memoryless wind-input-free SAC policy under the same current-observation setting. The resulting behavior, however, is also shaped by the reward function used to guide long-horizon sail–rudder control. To examine the role of different reward terms, a multi-seed reward-component ablation was conducted for SAC-LSTM under randomized wind conditions.

In each ablation case, one reward component was removed while the SAC-LSTM architecture, wind-input-free observation setting, training procedure, and evaluation protocol were kept unchanged. The progress reward provides dense navigation feedback, the action penalty regularizes sail–rudder variation, and the checkpoint and arrival rewards provide sparse task-level shaping. This ablation is used to examine whether each term primarily affects task completion, control smoothness, or navigation-efficiency shaping. The results are summarized in Table VI.

The ablation results in Table VI show that the reward components affect different aspects of the learned SAC-LSTM behavior. Removing the progress reward caused the most severe degradation: the task success rate decreased from 99.36 ± 0.26% to 1.76 ± 0.74%, and the final distance increased from 10.29 ± 1.18 m to 658.17 ± 5.21 m. This indicates that dense progress feedback is the primary learning signal for long-horizon wind-input-free navigation. Sparse checkpoint and arrival rewards alone are therefore insufficient to guide reliable sail–rudder control under randomized wind conditions.

The action penalty mainly affects control smoothness rather than task completion. Without this term, SAC-LSTM still maintained a high success rate of 98.48 ± 0.33%, but the sail-angle variation rate increased from 0.875 ± 0.008°/s to 1.222 ± 0.018°/s, and the rudder-angle variation rate increased from 0.633 ± 0.013°/s to 1.148 ± 0.011°/s. This result suggests that action regularization suppresses excessive sail–rudder adjustments after feasible navigation behavior has been learned.

Removing the checkpoint reward preserved a high success rate but increased the successful-episode average time consumption and trajectory length, suggesting that checkpoint-based shaping mainly improves navigation efficiency rather than serving as an essential completion signal. Removing the arrival reward had the weakest effect on the main randomized-wind metrics among the tested components. This does not imply that the arrival reward is generally unnecessary; rather, under the present reward formulation, dense progress feedback and checkpoint shaping already provide most of the guidance needed for target reaching.

Overall, Table VI suggests that the progress reward is the dominant navigation-learning component, the action penalty is the main smoothness-related component, and the checkpoint and arrival rewards provide secondary shaping under the tested randomized-wind setting.

4. Discussion

4.1. Recurrent temporal context under wind-input-free control

The main finding of this study is that recurrent observation–action history improves wind-input-free sail–rudder control under the tested partially observable sailing conditions. We attribute this improvement primarily to the use of recurrent history, because SAC-nowind and SAC-LSTM share the same wind-input-free current observation and differ mainly in whether recent observation–action information is encoded. SAC-nowind makes decisions from the current onboard vessel state and target-relative information, whereas SAC-LSTM additionally uses recent vessel-motion evolution and previous sail–rudder commands. The

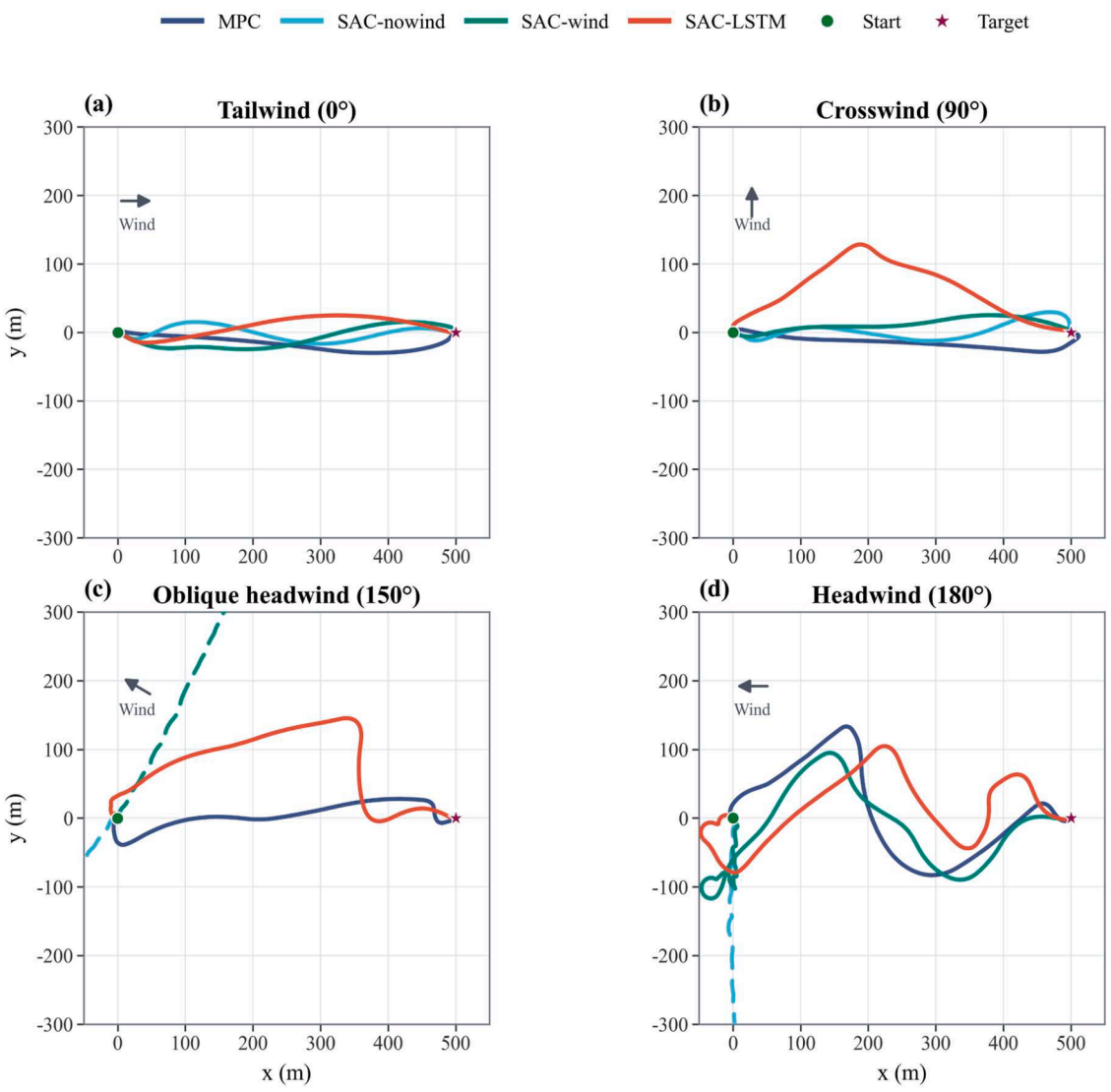


Fig. 11. Selected trajectory comparisons under four representative wind directions.

Table VI
Ulti-seed reward-component ablation of SAC-LSTM under randomized wind conditions.

Metric	Full SAC-LSTM	w/o $R_{progress,t}$	w/o $R_{action,t}$	w/o $R_{checkpoint,t}$	w/o $R_{arrival,t}$
Task success rate(%)	99.36 ± 0.26	1.76 ± 0.74	98.48 ± 0.33	99.56 ± 0.33	99.88 ± 0.11
	[99.04, 99.68]	[0.84, 2.68]	[98.06, 98.90]	[99.15, 99.97]	[99.74, 100.00]
Average time consumption(s)	177.22 ± 1.61	390.30 ± 28.26	179.81 ± 3.07	194.37 ± 2.39	176.75 ± 1.69
	[175.23, 179.22]	[355.20, 425.39]	[176.00, 183.63]	[191.41, 197.34]	[174.65, 178.86]
Average trajectory length(m)	586.38 ± 4.15	671.54 ± 75.97	572.03 ± 4.62	639.58 ± 5.40	594.38 ± 4.58
	[581.24, 591.53]	[577.22, 765.87]	[566.30, 577.76]	[632.87, 646.29]	[588.69, 600.07]
Sail angle variation rate(°/s)	0.875 ± 0.008	0.031 ± 0.006	1.222 ± 0.018	0.869 ± 0.012	0.825 ± 0.011
	[0.866, 0.885]	[0.023, 0.038]	[1.200, 1.244]	[0.855, 0.884]	[0.812, 0.839]
Rudder angle variation rate(°/s)	0.633 ± 0.013	0.024 ± 0.005	1.148 ± 0.011	0.678 ± 0.015	0.583 ± 0.007
	[0.617, 0.649]	[0.018, 0.029]	[1.135, 1.161]	[0.660, 0.697]	[0.574, 0.591]
Final distance (m)	10.29 ± 1.18	658.17 ± 5.21	13.61 ± 1.65	10.13 ± 0.89	8.63 ± 0.25
	[8.82, 11.75]	[651.70, 664.63]	[11.57, 15.65]	[9.03, 11.24]	[8.32, 8.93]
Total reward	404.33 ± 2.26	−279.99 ± 7.60	365.77 ± 1.64	403.79 ± 1.81	412.07 ± 0.34
	[401.53, 407.13]	[−289.42, −270.55]	[363.73, 367.80]	[401.54, 406.03]	[411.65, 412.49]

Remark: Values are reported as mean ± between-seed SD [95% CI] across five seed-level estimates from independently trained policies for each reward configuration. The 95% CIs are two-sided Student's t intervals with four degrees of freedom. For task success rate, interval endpoints outside [0%, 100%] are truncated to the admissible range for display. Task success rate, final distance, and total reward are calculated over all evaluation episodes; the remaining metrics are calculated over successful episodes.

improvement of SAC-LSTM therefore indicates that temporal context reduces part of the decision ambiguity caused by hidden wind

conditions.

This interpretation is consistent with the physical structure of sailing-vessel control. Wind affects the vessel through the coupled evolution of heading, velocity, roll motion, sail angle, and rudder response. When wind speed and wind direction are excluded from the policy input, their effects can still appear indirectly in how the vessel responds after recent control actions. The recurrent branch does not explicitly reconstruct the wind field. Instead, it gives the policy a short-term memory of vessel response, allowing sail–rudder commands to depend on both the current observation and recent dynamic evolution.

To place the proposed policy in a broader control context, SAC-wind and MPC are considered as references with different information structures. SAC-wind receives noisy wind measurements and uses wind-direction-related reward guidance, representing a wind-informed learning approach under the implementation used in this study. Its high completion rates in the tailwind and crosswind cases demonstrate the usefulness of direct wind information under these conditions. In the more demanding headwind-related cases, however, its lower completion rates indicate that instantaneous noisy wind measurements and wind-direction-related guidance did not provide equally robust performance within the present feedforward SAC implementation. This result is specific to the controller formulation and experimental settings considered here and does not characterize wind-informed control methods in general. MPC, in turn, uses explicit wind information, the vessel model, and online finite-horizon optimization, and therefore provides a model-based reference for the representative-wind trajectory behavior. Together, these comparisons position the wind-input-free recurrent policy relative to wind-informed learning and model-based control, while the comparison between SAC-nowind and SAC-LSTM identifies the contribution of recurrent history under the same wind-input-free current-observation setting.

The ablation results indicate that temporal representation and reward design play complementary roles. Recurrent history supplies decision context under hidden wind conditions, whereas the reward components shape the balance among task completion, navigation efficiency, and control smoothness. In particular, dense progress feedback provides the principal navigation-learning signal, while action regularization primarily limits excessive control variation.

Overall, the results support an application-specific conclusion: recent observation–action history can provide useful temporal context for wind-input-free sail–rudder control under the tested 4-DOF still-water simulation conditions. The contribution is therefore complementary to wind-aware GNC, MPC, and other model-based control methods. It identifies a data-driven wind-input-free control formulation in which recent observation–action history serves as a practical source of temporal context when direct wind variables are unavailable, unreliable, or intentionally excluded from the online policy input.

4.2. Limitations

This study examines whether recurrent observation–action history can support wind-input-free sail–rudder control under a controlled partially observable simulation setting.

The first limitation concerns simulation and model fidelity. The experiments were conducted in a control-oriented 4-DOF still-water simulation, with wind treated as the main hidden environmental factor affecting sail-generated propulsion. The simulation does not include heave and pitch motion, wave-induced loads, ocean currents, coupled environmental disturbances, or long-duration offshore degradation effects. Real sailing vessels may also exhibit hydrodynamic, aerodynamic, structural, actuator, and sensor characteristics beyond those represented in the present model.

The second limitation concerns generalization and sensitivity assessment before application-oriented validation. The principal evaluation used a fixed vessel parameterization, fixed task scale, selected wind settings, and the Dryden wind-field model described in Section 3.2. The policy was not systematically tested across different vessel sizes, sail configurations, actuator limits, target distances, wind-speed ranges, sea states, environmental models, or unseen disturbance distributions. The recurrent architecture, recurrent training sequence length, and main training hyperparameters were also fixed in the main experiments. A broader multi-seed evaluation of alternative recurrent unroll lengths remains for future work.

The third limitation concerns deployment-oriented validation. The nominal Gaussian observation noise used in this study represents moderate perturbations of selected onboard states, but it does not cover long-term sensor drift, intermittent measurement loss, calibration errors, corrupted measurements, sensor failure, actuator faults, or delayed onboard feedback. The present evaluation also does not include hardware-in-the-loop experiments, field trials, sim-to-real transfer tests, or dedicated onboard-feasibility benchmarks. Onboard feasibility was not separately assessed in terms of inference latency, memory usage, embedded-hardware constraints, or long-duration online reliability.

Overall, the controlled scope of this study was used to keep the recurrent-history effect experimentally identifiable while allowing large-scale reinforcement-learning training and seed-level statistical evaluation to remain computationally tractable. Within this scope, the results provide focused evidence for the value of recurrent observation–action history in wind-input-free sail–rudder control. Future work can build on this basis by incorporating higher-fidelity coupled marine simulations, broader vessel and environmental variations, degradation-aware sensing and actuation tests, embedded inference evaluation, and controlled field-oriented validation.

5. Conclusion

This study examined wind-input-free sail–rudder control for unmanned sailing vessels under partial observability. A recurrent SAC-LSTM policy was used to encode recent observation–action history together with onboard observable vessel states and target-relative information, while measured wind speed and wind direction were excluded from the policy input. The method was evaluated in a 4-DOF still-water simulation with a time-correlated Dryden wind field, observation noise, and five independent training seeds.

The main finding is that recurrent observation–action history improves wind-input-free control relative to the memoryless SAC baseline under the same current-observation setting. In the randomized-wind evaluation, SAC-LSTM achieved a task success rate of 99.36%, compared with 39.64% for SAC-nowind and 95.24% for the wind-informed SAC reference. The representative-wind results further showed that recurrent memory was particularly useful in oblique-headwind and headwind cases, where the memoryless wind-input-free policy failed to complete the task.

These results indicate that recent vessel-motion responses and previous sail–rudder actions can provide useful temporal context for sail–rudder decision-making when direct wind variables are unavailable to the online policy. Taken together, these findings position recurrent observation–action history as a useful temporal information source for limited-sensing sail–rudder control, and support SAC-LSTM as a complementary data-driven control option under the tested simulation conditions.

Data and code availability statement

The source code, configuration files, training and evaluation scripts, and representative trained checkpoints are available in a public GitHub repository, release v1.0.3: https://github.com/Sailboat-RL-4DOF-Control/sailboat2026/releases/tag/v1.0.3. The seed-level evaluation summaries, extended descriptive statistics, and seed-level statistical comparison tables supporting the main results are provided in the Supplementary Material.

CRediT authorship contribution statement

Chao Yang: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation. **Buyu Guo:** Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization. **Yulu Zhang:** Validation, Software, Resources, Data curation. **Yanzhen Gu:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Yanjun Li:** Writing – review & editing, Resources, Formal analysis, Data curation. **Sheng Huang:** Visualization, Software, Data curation. **Hui Huang:** Visualization, Supervision, Formal analysis, Data curation. **Jiang Dong:** Visualization, Resources, Investigation, Data curation. **Peiliang Li:** Supervision, Project administration, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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